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BACHELOR'S THESIS

DESIGN AND IMPLEMENTATION OF A REDUNDANT DATA CENTER-CAMPUS NETWORK USING CISCO PACKET TRACER

Candidate: Anton-Alexandru Demian

Scientific Supervisor: Assoc. Prof. Dr. Eng. Janos Gal

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ABSTRACT

This project demonstrates how a redundant Data Center-Campus network infrastructure can be designed, configured, and tested using Cisco Packet Tracer. The main objective of the thesis is to build a segmented and secure network capable of maintaining connectivity during simulated failures.

The network topology is designed and tested in Cisco Packet Tracer, which enables the creation of a virtual infrastructure consisting of routers, switches, servers, and end stations. The network is organized into several functional areas, including the Routing Layer, Core / Distribution, Campus Distribution, Campus Access, Data Center, and DMZ. This organization enables the separation of users, servers, and services that are exposed in a controlled manner.

Chapter 1 presents the development of modern telecommunications networks and the evolution of network infrastructure requirements. It highlights the need for segmentation, availability, and security within an organizational network.

Chapter 2 describes the theoretical foundations required for the project. It covers Campus and Data Center network concepts, VLAN segmentation, inter-VLAN routing, the OSPF protocol, redundancy mechanisms such as HSRP, Rapid STP, and EtherChannel, as well as the role of ACLs and network services.

Chapter 3 presents the infrastructure design. It defines the project requirements, the Data Center-Campus architecture, the functional areas, the equipment used, the VLAN plan, the IP addressing scheme, the security policies, and the redundancy strategy.

Chapter 4 details the implementation and configuration of the network in Cisco Packet Tracer. It covers the configuration of Layer 3 devices, VLANs and trunk links, VLAN interfaces, inter-VLAN routing, HSRP, OSPF, Layer 2 mechanisms, network services, and ACL policies.

Chapter 5 presents the testing and validation of the infrastructure. Basic connectivity, access to Web, DNS, FTP, and E-mail services, the enforcement of security policies, and the operation of redundancy mechanisms are verified. Failover tests for HSRP, OSPF, EtherChannel, and Rapid STP confirm that the network can maintain communication when the state of links or devices changes.

Chapter 6 presents the project conclusions, highlighting that the implemented topology meets the proposed objectives: logical segmentation, routing between functional areas, controlled access to services, DMZ isolation, and redundancy at both Layer 2 and Layer 3.

Chapter 7 contains the bibliographic references used in preparing the thesis.

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1. INTRODUCTION

1.1 DEVELOPMENT OF MODERN TELECOMMUNICATIONS NETWORKS

Local area networks emerged from the need to connect devices located in the same site and to provide shared access to resources and services. Over time, their role expanded from simply connecting devices to providing an organized infrastructure capable of supporting stable communication between users and servers. Even in a small or medium-sized network, traffic must be properly structured and access to resources must be controlled according to the role of each network area.

An important step in the development of networks was the transition from simple broadcast domains to logically segmented networks. In an unsegmented network, all devices may belong to the same broadcast domain, which can lead to unnecessary traffic, management difficulties, and security risks. VLANs were introduced to address these issues by logically dividing a network into multiple independent segments. By using VLANs, devices can be grouped according to role, department, or access level without requiring physical separation of the infrastructure.

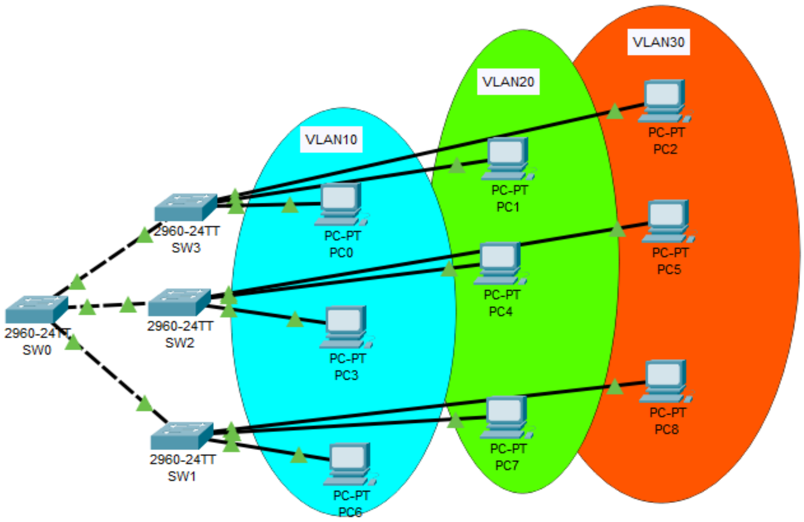

Figure 1.1.1 - Logical network segmentation using VLANs

The hierarchical design model divides the network into functional layers such as access, distribution, and core. The access layer connects end-user devices, the distribution layer aggregates traffic and applies control policies, while the core layer provides fast data transport between the main areas of the network. This structure simplifies administration, improves troubleshooting, and provides greater flexibility for future network expansion.

Alongside the development of campus networks, data centers have become increasingly important. A Data Center is the infrastructure in which applications, servers, storage systems, and services that are critical to an organization's operation are hosted. It is not simply a room containing servers, but an area designed for availability, security, and performance. A Data Center includes equipment such as servers, switches, routers, storage systems, and other components required to deliver digital services to users.

A modern infrastructure must do more than simply connect devices; it must continue to operate reliably even when failures occur. For this reason, redundancy has become an important principle in network design. If access to a critical resource depends on a single device or link, failure of that component can interrupt the service. To avoid such situations, modern networks use redundant devices, multiple links, dynamic routing protocols, and failover mechanisms. These solutions allow the network to continue operating even when one component becomes unavailable. [1] [2]

1.2 EVOLUTION OF NETWORK INFRASTRUCTURE REQUIREMENTS

As telecommunications networks have evolved, the requirements placed on network infrastructures have become increasingly complex. While the primary purpose of early networks was to connect devices and share resources, a modern infrastructure must provide stable connectivity, controlled access to services, high availability, and security.

In organizational environments, the network no longer serves only isolated workstations; it also supports applications, servers, communication services, file transfers, web services, and access to internal resources. For this reason, a network must be designed according to clear principles so that traffic is organized and different categories of users and services can be managed efficiently.

An important requirement of modern infrastructures is the logical separation of resources. Standard users, management devices, internal servers, and services exposed in a controlled manner should not all be treated in the same way. Each area of the network may have a different role and, therefore, a different level of access. This separation improves manageability and reduces security risks.

At the same time, service availability has become an essential requirement. A link failure or device failure can affect users' access to important applications and services. To reduce dependence on a single critical point, modern networks use redundant links, backup devices, and protocols capable of maintaining connectivity when failures occur.

Security is also an important component of network design. Access to servers, management devices, and sensitive areas must be controlled through clear policies. A well-designed infrastructure must allow the communications required for services to operate while limiting unauthorized or unnecessary traffic between network segments.

In this context, Campus and Data Center architectures provide a suitable model for organizing a modern infrastructure. The Campus area can serve users and end devices, while the Data Center area can host the organization's important servers and services. Separating these areas produces a clearer design in which traffic, access, and security can be controlled more effectively.

2. THEORETICAL FOUNDATIONS

2.1 CAMPUS AND DATA CENTER NETWORKS

In modern communications infrastructures, networks are designed according to the role they perform within an organization. Two essential concepts in this field are Campus networks and Data Center networks. These are not merely separate physical areas, but functional components of the same infrastructure, each with specific responsibilities and objectives.

A Campus network is the infrastructure that connects users and end devices to an organization's services. This category includes workstations, laptops, printers, IP phones, wireless access points, and other devices used in day-to-day operations. According to Cisco's Campus LAN and Wireless LAN design guide, the term "campus" is not tied to a fixed geographic size and can describe both a simple network based on a single switch and an infrastructure spanning multiple buildings [2].

The primary purpose of a Campus network is to provide access to organizational resources in a secure, stable, and efficient manner. To achieve this objective, the infrastructure must be designed to organize traffic, separate different categories of users, and apply appropriate access policies. Devices and users within an organization have different roles, which creates the need for distinct access levels. For example, management devices, administrative users, and standard users can be logically separated to improve security and simplify network administration.

From an architectural perspective, Campus networks are often designed according to a hierarchical model. This model divides the infrastructure into several functional layers: Access Layer, Distribution Layer, and Core Layer. The Access Layer connects end devices, the Distribution Layer aggregates traffic from access switches and applies control policies, while the Core Layer provides fast and efficient transport between the main network segments. Cisco emphasizes that this modular structure simplifies infrastructure design, administration, and troubleshooting [2].

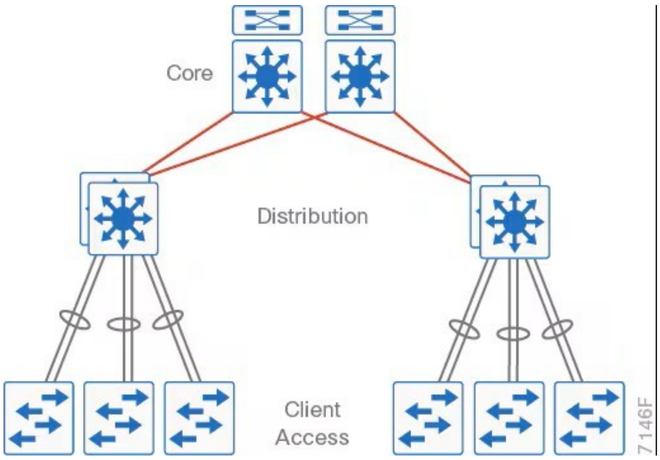

Figure 2.1.1 - Hierarchical architecture of a Campus network based on the Access, Distribution, and Core layers (source: https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html)

As shown in the figure above, the hierarchical model separates network functions across multiple layers. The access layer connects users and end devices, the distribution layer aggregates traffic and applies control policies, while the core layer provides fast data transport between the main infrastructure segments. This type of organization supports a scalable network that is easier to manage and more resilient to failures.

2.1.1 CISCO HIERARCHICAL MODEL

The Cisco hierarchical model is used to organize Campus networks into distinct functional layers. This approach separates responsibilities within the infrastructure and contributes to a clearer, more scalable, and easier-to-manage design. Instead of connecting all devices in a flat structure, the hierarchical model divides the network into layers, each with a clearly defined role.

The three main layers of the model are Access, Distribution, and Core. The Access Layer connects end devices such as computers, printers, IP phones, and wireless access points. The Distribution Layer aggregates traffic from the Access Layer and can apply routing, filtering, and access-control policies. The Core Layer is responsible for rapidly transporting traffic between the main network areas and is designed for high performance and availability [2].

Table 2.1.1.1. Roles of the layers in the Cisco hierarchical model

Layer	Main function	Example roles in the network
Access	Connecting end devices	Workstations, laptops, printers, IP phones, access points
Distribution	Traffic aggregation and policy enforcement	Inter-VLAN routing, ACLs, security policies, redundancy
Core	Fast transport between the main areas	High-speed connectivity, redundancy, transport between important segments

Using this model makes the network design more modular. Each layer can be analyzed, expanded, or troubleshot independently without directly affecting the entire infrastructure. The model also allows redundancy and security mechanisms to be introduced at the appropriate points in the network. For example, access-control policies can be applied at the Distribution Layer, while the Core Layer can be optimized for fast transport and stability.

Unlike the Campus network, which is user-oriented, the Data Center is focused on delivering services and applications. Cisco defines a Data Center as a physical facility used to host an organization's applications and critical data. It is built around an infrastructure that integrates computing, storage, and communications resources required to deliver services to users [3].

A Data Center contains servers, switches, routers, firewalls, storage systems, and other equipment essential to the operation of digital services. Hosted services can include web applications, DNS services, e-mail, FTP, databases, and various internal applications. These services must remain available to users while also being protected through appropriate security and access-control mechanisms.

The interconnection between the Campus network and the Data Center is essential to the organization's operation because users in the Campus area must be able to access services hosted in the Data Center. However, access to these resources must be properly managed and controlled. In a well-designed infrastructure, traffic between users and servers is organized through logical segmentation, routing, and filtering mechanisms. As a result, different categories of users can be assigned different levels of access to internal resources.

2.2 NETWORK SEGMENTATION AND TRAFFIC ROUTING

In a modern network infrastructure, connecting devices alone is not enough to ensure efficient and secure operation. Traffic must be organized according to the roles of the devices, users, and services present in the network. For this reason, logical segmentation and traffic routing are two important components in the design of a Campus-Data Center network.

Logical segmentation is commonly implemented using VLANs. A VLAN divides a physical infrastructure into multiple separate logical networks. Devices can therefore be grouped according to role, department, or access level without requiring separate physical equipment. For example, management devices, administrative users, standard users, and servers can belong to different VLANs. This organization reduces unnecessary traffic, simplifies administration, and improves security.

In an unsegmented network, all devices may reside in the same broadcast domain, which can generate additional traffic and make access control more difficult. By using VLANs, each category of device can be logically isolated. This separation is especially important in an infrastructure containing workstations, internal servers, management devices, and services that should be accessible only to specific categories of users.

Trunk links are used to carry VLANs between switches. A trunk link allows traffic from multiple VLANs to be transmitted over the same physical connection between devices. The IEEE 802.1Q standard identifies VLAN traffic on trunk ports so that switches can separate traffic according to the VLAN to which it belongs [4]. In this way, the same physical infrastructure can support multiple separate logical networks.

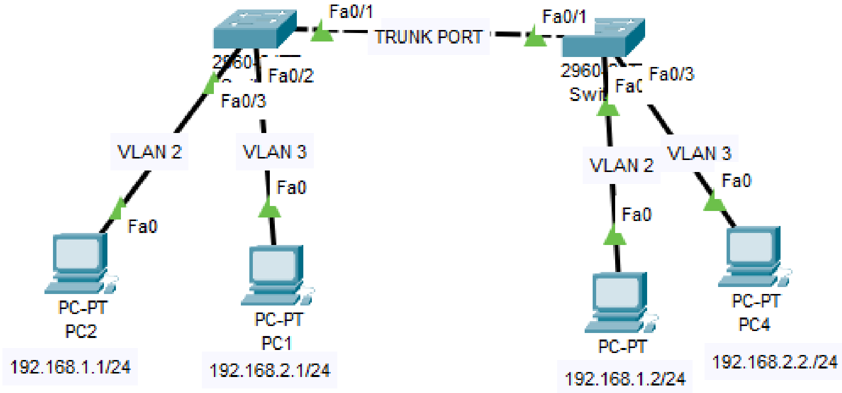

Figure 2.2.1 - Example of a trunk link between two switches carrying traffic for multiple VLANs.

The figure illustrates a simple topology in which two switches are connected by a trunk link. End stations belong to different VLANs, and the link between the switches carries traffic for multiple VLANs over the same physical connection. This solution is useful when the same VLANs need to span multiple switches. However, VLAN-based segmentation also creates the need for routing between these segments whenever communication between them is required. Devices in different VLANs cannot communicate directly at Layer 2 because they belong to separate logical networks. For traffic to pass between VLANs, a Layer 3 device such as a router or multilayer switch must perform inter-VLAN routing.

Inter-VLAN routing can be implemented using a router or a Layer 3 switch. Layer 3 switches are commonly used in modern networks because they can perform routing directly within the switching infrastructure. They use virtual interfaces known as SVIs, with one SVI for each routable VLAN. Each SVI has an IP address and can operate as the gateway for devices in the corresponding VLAN [5].

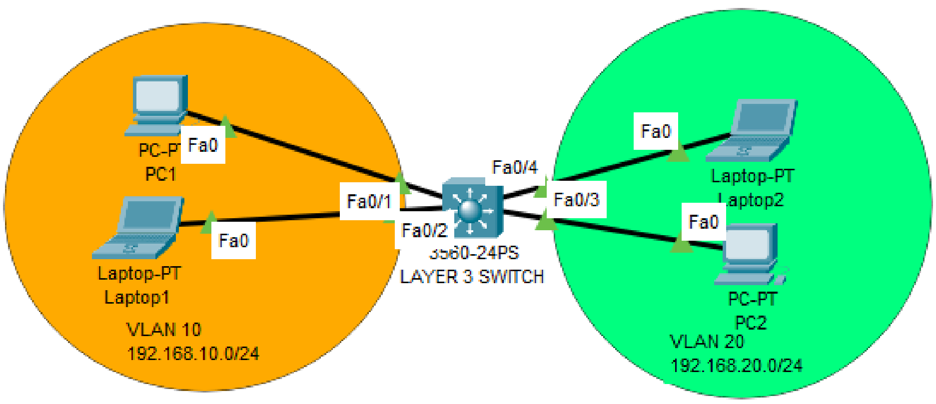

Figure 2.2.2 - Example of inter-VLAN communication through a Layer 3 switch.

Inter-VLAN routing makes it possible to organize and control communication between network areas. For example, users in one VLAN may be allowed to access a web server while being restricted from accessing an FTP server or management devices. Routing therefore serves not only to enable communication between networks, but also to provide a point at which control and security policies can be applied.

In a Campus-Data Center infrastructure, routing plays an important role in connecting the user area to the server area. The Campus network is oriented toward end devices, while the Data Center hosts services such as Web, DNS, FTP, and E-mail. For these services to be accessible to users, communication between network segments is required, but that communication must be properly organized, routed, and filtered.

In addition to inter-VLAN routing, modern networks use dynamic routing protocols. One example is OSPF, a link-state routing protocol used in IP networks and classified as an Interior Gateway Protocol. OSPF is suitable for use within a single autonomous system, for example in an enterprise network.

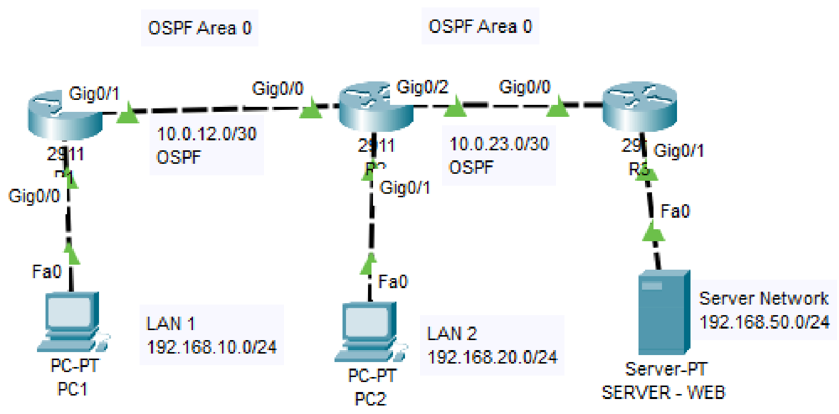

Figure 2.2.3 - Example of dynamic OSPF routing in a three-router topology.

During OSPF operation, devices exchange information about the network topology through LSA messages, build a link-state database, and calculate the best routes using the Shortest Path First algorithm. Therefore, when a topology change occurs, such as a link failure, the protocol can recalculate routes and use an alternative path if one is available [6].

2.2.1 ADVANTAGES OF USING OSPF

Using OSPF in a network infrastructure provides several advantages over manually configured static routes. One major advantage is the protocol's ability to adapt automatically to topology changes. If a link becomes unavailable, routers participating in OSPF can recalculate their paths and use an alternative route when one exists.

Another important advantage is fast convergence. OSPF is a link-state protocol, which means that devices exchange information about the state of network links and build a logical view of the topology. Based on this information, each device calculates the best routes using the Shortest Path First algorithm. As a result, the network can react to changes more efficiently than solutions based exclusively on static routes [6].

OSPF also supports VLSM, allowing variable-length subnet masks to be used. This is important in modern networks because it enables more efficient use of the IP address space. In an infrastructure containing multiple VLANs, point-to-point links, and different functional areas, using masks tailored to each segment supports more efficient address planning [6].

Scalability is another reason OSPF is used in enterprise networks. The protocol can operate in larger topologies and organize routing into areas, allowing routing information to be managed more efficiently.

VLAN segmentation and traffic routing are therefore complementary mechanisms. VLANs logically separate network areas, trunk links carry those VLANs between switches, inter-VLAN routing enables controlled communication between segments, and OSPF provides flexibility in route exchange between Layer 3 devices. Together, these concepts contribute to an organized and scalable network suitable for interconnecting a Campus area with a Data Center area.

2.3 REDUNDANCY, AVAILABILITY, AND ACCESS CONTROL

Availability is an important requirement when designing a network infrastructure. In an organization, a loss of connectivity between users and internal services can affect application access, communication, and day-to-day operations. Modern networks are therefore designed not to depend on a single link or a single device. Redundancy provides alternative paths that can be used if a network component becomes unavailable.

Redundancy can be implemented at multiple levels. At the physical level, multiple links can exist between devices. At the infrastructure level, redundant routers and switches can be used. At the logical level, protocols can detect network changes and automatically select a working path. The purpose of these mechanisms is to maintain connectivity and reduce downtime when a failure occurs.

In Layer 2 networks, using multiple links between switches can create switching loops. These loops can generate excessive traffic, instability, and operational problems throughout the network. Spanning Tree Protocol and its modern variants, such as Rapid Spanning Tree Protocol, are used to prevent these situations. STP allows redundant links to exist between switches without creating Layer 2 loops.

STP operates by selecting a loop-free logical topology. When multiple paths exist between switches, the protocol can temporarily block selected ports while keeping them available as backup paths. If an active link becomes unavailable, the protocol can recalculate the topology and activate an alternative path. Rapid STP improves this process by providing faster convergence, which is useful in networks where recovery time after a failure must be reduced [7].

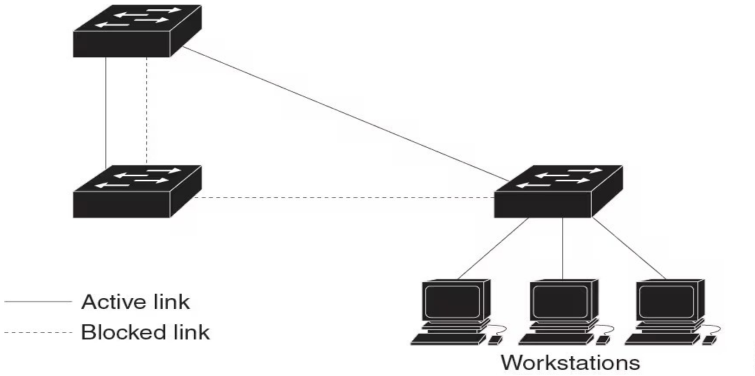

Figure 2.3.1 - Example of a Layer 2 topology in which STP blocks a redundant link to prevent loops. (source: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9200/software/release/17-14/configuration_guide/lyr2/b_1714_lyr2_9200_cg/configuring_spanning_tree_protocol.html)

The figure illustrates how STP maintains a loop-free logical topology in a network with redundant links. Solid lines represent active links, while dashed lines represent links temporarily blocked by the protocol. If an active link becomes unavailable, STP can reactivate a backup path, thereby helping maintain network connectivity.

Another mechanism used to increase availability and performance is EtherChannel. It allows multiple physical links to be grouped into a single logical link called a port-channel. From the devices' perspective, multiple cables operate as a single logical connection. If one physical link fails, traffic can continue to flow through the remaining active links [8].

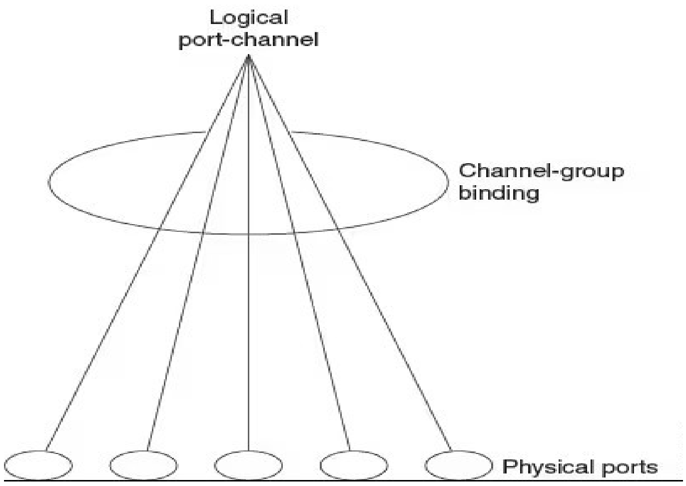

Figure 2.3.2 - Relationship between physical ports, the EtherChannel group, and the logical port-channel interface. (source: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9200/software/release/17-14/configuration_guide/lyr2/b_1714_lyr2_9200_cg/configuring_etherchannels.html)

Figure 2.3.2 highlights that EtherChannel groups multiple physical ports into a single logical interface called a port-channel. Functionally, the devices treat this group of links as one connection, which both increases availability and enables more efficient use of bandwidth. If one of the physical links in the group becomes unavailable, traffic can continue to flow through the remaining active ports.

Another important element in a redundant network is the gateway used by end stations. Normally, each station is configured with a default gateway address through which it communicates outside its local network. If the device acting as the gateway becomes unavailable, users may lose access to other networks even if the rest of the infrastructure is still operating. First Hop Redundancy Protocols are used to avoid this problem.

HSRP is an example of a First Hop Redundancy Protocol. It allows multiple Layer 3 devices to provide a shared virtual gateway for end stations. One device operates in the active role and responds for the gateway, while another remains in standby and is ready to take over if a failure occurs. By using HSRP, end stations can continue to use the same virtual gateway address without requiring configuration changes when the active device changes [9].

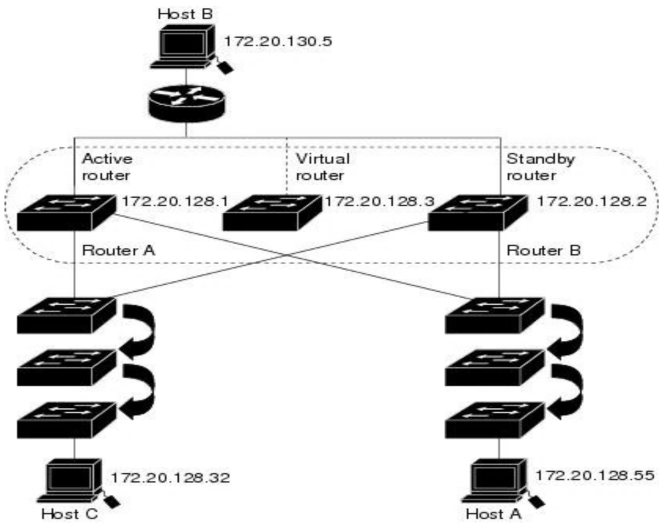

Figure 2.3.3 - HSRP operating principle (source: https://www.cisco.com/c/en/us/td/docs/switches/lan/c9000/lyr3-fwd/fhrp/fhrp-configuration-guide/hsrp.html)

HSRP is not the only protocol in the First Hop Redundancy Protocol category. Several solutions are designed to provide redundancy for the default gateway used by end stations. The best known are HSRP, VRRP, and GLBP. These protocols share the same general objective - maintaining gateway access when the primary device becomes unavailable - but differ in their operating model and level of standardization.

Table 2.3.3.1. Comparison of gateway redundancy protocols

Protocol	Vendor	Main characteristic
HSRP	Cisco	Creates a shared virtual gateway with one active and one standby device
VRRP	Open standard	Allows multiple routers to provide a shared virtual router for stations in the same network
GLBP	Cisco	Provides gateway redundancy and can distribute traffic across multiple devices

HSRP was selected for this thesis because the project is implemented in Cisco Packet Tracer and uses Cisco devices. HSRP is also well suited to the proposed scenario because it allows a stable virtual gateway to be configured for each VLAN. End stations can therefore use the same gateway address, and if the active Layer 3 switch becomes unavailable, the standby device can assume the role.

HSRP provides a shared virtual gateway for network stations so that they do not depend on a single physical device. Within an HSRP group, one router is designated as active and processes traffic destined for the virtual gateway, while a second router remains in standby and automatically takes over if the active device becomes unavailable. This approach maintains communication continuity and reduces the risk of losing access to network resources.

In addition to availability, security is an important requirement in network infrastructure design. Not every device should have access to every resource. Standard users, administrators, management devices, internal servers, and servers exposed in a controlled manner must be treated differently. Access can be controlled through Access Control Lists, commonly known as ACLs.

ACLs are rules applied to network traffic to determine which packets are permitted and which are blocked. These rules can be defined according to criteria such as the source IP address, destination IP address, protocol, or transport-layer port. By using ACLs, an administrator can control communication between different network areas and restrict access to sensitive resources [10].

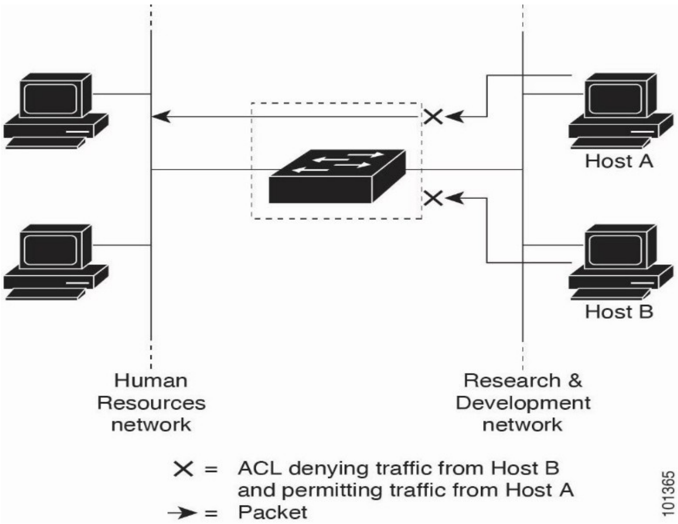

Figure 2.3.4 - Example of traffic filtering with an ACL, allowing traffic from one host while blocking traffic from another. (source: https://www.cisco.com/c/en/us/td/docs/switches/lan/c9000/security/acls/acls-configuration-guide/access-control-lists.html)

Figure 2.3.4 illustrates how an access control list can filter traffic between two network areas. In the example shown, traffic from Host A is permitted while traffic from Host B is blocked. This principle is used to implement security policies in which selected devices or networks are allowed to access resources while others are restricted.

For example, standard users can be granted access to a web server or the e-mail service while being restricted from accessing the FTP server or management devices. At the same time, administrators can have broader access to internal servers, while servers in other areas can be isolated from the rest of the internal infrastructure. ACLs therefore support the implementation of clear security policies between VLANs and different functional areas.

Server isolation is closely related to segmentation and access control. Internal servers can be placed in a dedicated VLAN, while servers that must be accessible in a controlled manner can be placed in a DMZ. This separation reduces the risk that an issue in one area will directly affect the entire infrastructure. At the same time, ACLs can control traffic between users, internal servers, and the DMZ.

2.3.1 DMZ

A DMZ, short for Demilitarized Zone, is a network segment separated from the internal network and used to host services that must be accessible in a controlled manner from other areas of the infrastructure. The main purpose of a DMZ is to provide an additional layer of protection between the internal network and services that may be exposed to users or less trusted networks [15].

In a network architecture, internal servers should not be exposed directly to untrusted areas. If a server providing public or semi-public services were placed in the same area as internal servers, a security incident could more easily affect the entire infrastructure. By placing such services in a DMZ, access can be controlled more strictly and the internal network remains better protected.

The DMZ can host services that need to be accessible in a controlled manner, such as Web servers, public DNS servers, or e-mail gateways. These services can communicate with users or other network areas, but their access to the internal network must be limited. The DMZ therefore acts as an intermediate area separated from both internal users and critical Data Center servers.

Traffic between the DMZ and the rest of the infrastructure is controlled through security policies such as ACLs or firewall rules. These rules determine which traffic types are allowed and which communications are blocked. For example, users may be granted HTTP access to a Web server in the DMZ, while the DMZ server can be prevented from directly accessing internal servers or management devices.

In the Cisco Packet Tracer project, the DMZ is used to host a Web server that is separate from the internal servers. This arrangement distinguishes internal services located in the Data Center from services exposed in a controlled manner in the DMZ. By applying ACLs, traffic to and from the DMZ can be filtered so that unauthorized access to the internal network is limited.

Combining redundancy mechanisms with security policies results in a more stable and better-controlled infrastructure. STP and Rapid STP prevent loops in Layer 2 networks, EtherChannel provides redundant logical links, HSRP maintains gateway availability, and ACLs control access between areas. Together, these technologies contribute to the design of a network capable of providing connectivity, availability, and security.

2.4 NETWORK SERVICES AND THE ROLE OF SERVERS

In a network infrastructure, servers provide services to users, devices, and applications. While switching and routing provide traffic transport, network services are the resources actually accessed by users. In an organizational network, these can include access to web pages, domain-name resolution, file transfers, e-mail communication, and other internal applications.

The client-server model underpins many network services. In this model, a client initiates a request to a server, and the server responds by providing the requested resource. The client may be a workstation, laptop, or application, while the server is the device hosting the service. This model enables centralized resources and more efficient management of the services provided over the network.

Server placement is an important aspect of infrastructure design. Servers providing internal services such as Web, DNS, FTP, and E-mail can be grouped in a dedicated area so that access to them can be managed and controlled more easily. In an organized architecture, this area can be considered the Data Center because it hosts the network's primary services.

Separating servers from user stations allows clear access policies to be applied. Users in the Campus network should not have unrestricted access to every service, but only to the resources required for their role. For example, Web or E-mail access may be permitted for several user categories, while FTP or management-service access may be restricted.

For services that must be accessible in a controlled manner from different areas of the network, a separate area called a DMZ can be used. Servers placed in the DMZ are isolated from internal servers, reducing the risk that an issue in an exposed area will directly affect the internal infrastructure. This separation improves security and clearly distinguishes areas with different access levels.

2.4.1 WEB SERVICE

An important service in a modern infrastructure is the Web service. It enables content to be accessed through a browser using HTTP or its secure variants. HTTP is a client-server protocol used to retrieve web resources such as HTML pages, images, files, or other elements of a web application [11]. In an organizational network, a Web server can be used for internal applications, information pages, administrative interfaces, or services exposed to users in a controlled manner.

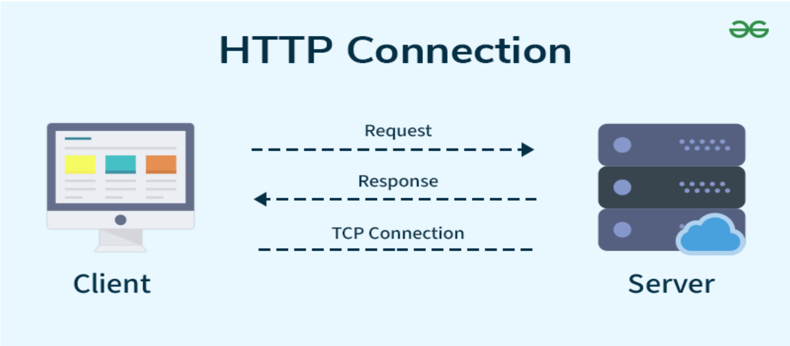

Figure 2.4.1.1 - Client-server model of an HTTP connection. (source: https://www.geeksforgeeks.org/blogs/http-full-form/)

Figure 2.4.1.1 illustrates the operating principle of an HTTP connection between a client and a server. The client, represented by a workstation using a browser, sends a request to the server for a web resource. The server processes the request and returns a response to the client, which can contain a web page, a file, or other elements required to display the content. This communication demonstrates the client-server model used to access Web services in a network infrastructure.

HTTP communication is based on a request-response mechanism. The client initiates the connection and requests a resource, while the server processes the request and returns an appropriate response. This response can indicate a successful operation, an unavailable resource, or an error at the server level.

2.4.2 DNS SERVICE

The operation of Web services is closely related to DNS. Users normally access servers through easy-to-remember domain names rather than directly using IP addresses. DNS translates these names into IP addresses so that network devices can identify the server to which traffic must be sent [12].

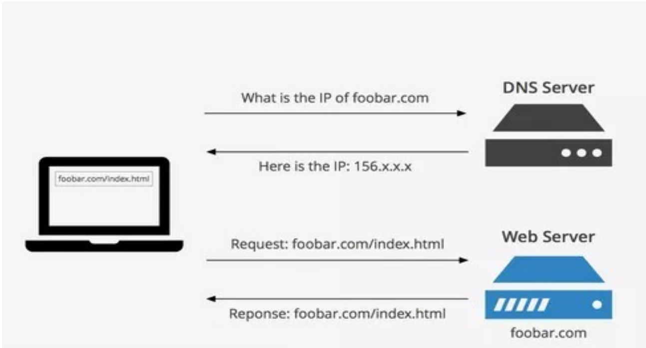

Figure 2.4.2.1 - DNS resolution process before accessing a Web server. (source: https://www.keycdn.com/support/what-is-a-dns-server)

Figure 2.4.2.1 illustrates the role of DNS when accessing a Web server. First, the client sends a request to the DNS server to obtain the IP address associated with a domain name. The DNS server returns the corresponding IP address, which the client can then use to initiate a connection to the Web server. This allows users to access services by using easy-to-remember names without having to memorize server IP addresses.

For example, when a user enters a name such as foobar.com in a browser, the client station does not initially know the IP address of the Web server. The client sends a request to the DNS server configured on the network, which looks up the corresponding record and returns the IP address associated with that name. Once the IP address has been obtained, the client can initiate the HTTP connection to the Web server.

Using DNS makes infrastructure administration simpler and more flexible. If a server's IP address changes, users can continue to use the same domain name, while only the DNS record needs to be updated. DNS therefore improves both service usability and the logical organization of network resources.

In an organizational infrastructure, DNS can be used for multiple types of services. In addition to Web servers, records can be defined for e-mail servers, FTP servers, or other internal applications. Users and network devices can therefore access services through clear names without having to memorize the IP address of each server.

2.4.3 FTP SERVICE

Another service used in network infrastructures is FTP, short for File Transfer Protocol. It is used to transfer files between a client and a server. Through FTP, a user can upload, download, or manage files stored on a server according to the configured access rights.

In an organizational network, an FTP server can be used to store and distribute documents, configuration files, archives, or other internal resources. Unlike the Web service, which is mainly oriented toward displaying content through a browser, FTP is designed for direct file transfer between devices.

FTP operates through communication between an FTP client and an FTP server. The client initiates a connection to the server, authenticates using a username and password, and can then perform file operations according to the granted permissions. According to the technical documentation, FTP uses one connection for commands and a separate connection for the actual data transfer [13].

The FTP service model can be represented by separating the control process, which handles user commands, from the data-transfer process, which is responsible for transferring files between the client and server.

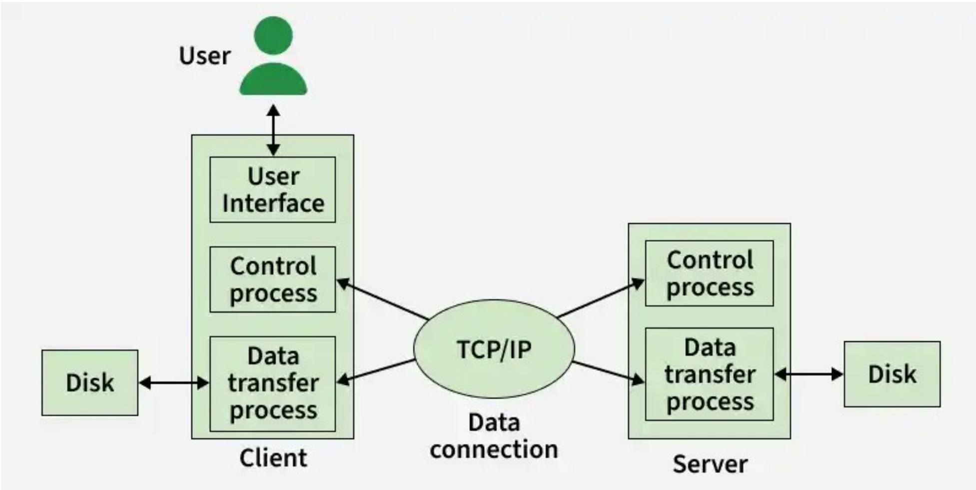

Figure 2.4.3.1 - Operating model of the FTP service between a client and a server. (source: https://www.geeksforgeeks.org/computer-networks/file-transfer-protocol-ftp-in-application-layer/)

Figure 2.4.3.1 illustrates how the FTP service transfers files between a client and a server. The user interacts with a client interface, and commands are sent to the server through a control process. The actual file transfer is performed separately through a dedicated data-transfer process. Both the client and server access their own storage systems, while communication between them takes place over the network using TCP/IP.

Because FTP provides access to files, the service must be carefully controlled. Not all user categories should have the same permissions on the FTP server. For example, an administrative user may have read and write permissions, while a standard user may have limited access or be completely restricted. In a segmented infrastructure, FTP access can be further controlled through security policies and access control lists.

2.4.4 E-MAIL SERVICE

The E-mail service enables messages to be sent and received between users over a network infrastructure. In an organization, e-mail is one of the most widely used communication services and is used to send information, notifications, documents, and internal messages.

The E-mail service also operates according to the client-server model. The user works with an e-mail application configured with an account and the address of the mail server. The client sends the message to the server, which then delivers it to the recipient. SMTP is the standard protocol used to transmit e-mail messages over a network [14].

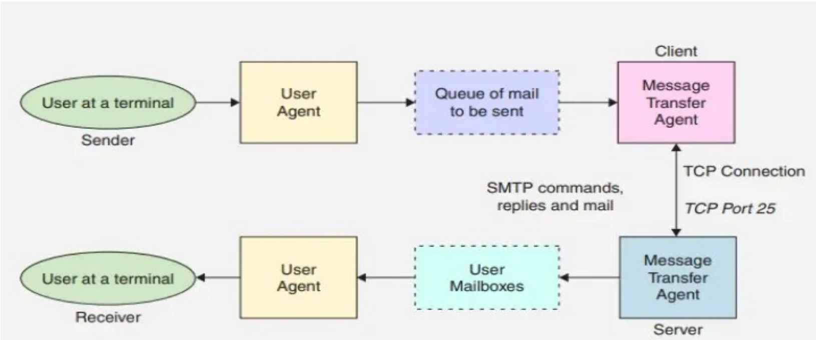

Figure 2.4.4.1 - Simplified model for transmitting a message through the E-mail service. (source: https://www.geeksforgeeks.org/computer-networks/simple-mail-transfer-protocol-smtp/)

Figure 2.4.4.1 illustrates the operating principle of the E-mail service. The user sends a message through an e-mail client, and the mail server processes and delivers the message to the recipient. In an internal infrastructure, this service can be used to test communication between users and verify access to the e-mail server from different network areas.

In a network infrastructure, the e-mail server must be configured with a domain, user accounts, and client access parameters. For example, an internal network may contain users such as admin, student, or management, each with their own e-mail address and authentication credentials. End stations must be configured with the sending and receiving server so that users can communicate with one another.

The E-mail service can also depend on DNS because the mail server can be accessed through a domain name rather than only by its IP address. Instead of configuring the server's numeric address directly, users can use a name such as mail.campus.com. The DNS server translates this name into the e-mail server's IP address, allowing the client to establish a connection to the appropriate service.

From a security perspective, access to the E-mail service must be controlled through authentication and network-level access policies. Not all network areas should have unrestricted access to the e-mail server, and traffic can be permitted or restricted according to user roles. In a segmented network, E-mail access can be allowed for selected VLANs and blocked for areas that do not require the service.

Servers should be placed in the network according to each service's role and the required level of exposure. Internal servers such as DNS, FTP, or E-mail can be located in a protected Data Center area. Services that must be accessible in a controlled manner from less trusted areas can be placed in a separate area such as a DMZ. This separation helps limit the impact of a potential incident and protects the internal infrastructure.

Network services should not be considered separately from segmentation, routing, and security mechanisms. For a user to access a server, connectivity must exist between the client network and the server network, routing between the segments must be correct, and access policies must permit the required traffic. At the same time, unnecessary traffic should be restricted to reduce security risks.

3. NETWORK INFRASTRUCTURE DESIGN

3.1 PROJECT REQUIREMENTS

The project requirements were defined based on the need to build a Data Center-Campus network that includes users, servers, routing and switching devices, logically separated areas, and traffic-protection mechanisms. The network must allow users to access the required services while limiting unauthorized communication toward sensitive areas of the infrastructure.

A first requirement is logical network segmentation. For this purpose, the infrastructure is divided into several VLANs corresponding to the roles present in the topology: Management, Admin, Students, Internal Servers, and DMZ. This separation organizes traffic according to user and service categories and reduces the risk of all devices operating within a single logical domain.

Another requirement is communication between VLANs. Because each VLAN represents a separate logical network, communication between users and servers must be performed by Layer 3 devices. In this project, that function is provided by the Layer 3 switches, which route traffic between subnets and act as central points for traffic control.

Network availability is an important requirement because the infrastructure should not depend on a single device or a single link. For this reason, the topology includes two routers, two Layer 3 switches, redundant links between devices, and protocols capable of maintaining connectivity in the event of a failure. Redundancy is implemented through HSRP, OSPF, Rapid STP, and EtherChannel.

Traffic security is another essential requirement. The network must allow users to access the services they need while blocking communications that are not justified. For example, users in the Students VLAN should not have access to management devices or to all internal services. At the same time, the DMZ must be isolated from the internal network to reduce the risk of unauthorized access to critical servers.

The project must also include functional network services. The Data Center contains servers for Web, DNS, FTP, and E-mail, while the DMZ contains a separate Web server. These services are used to test connectivity, domain-name resolution, file transfer, and e-mail communication.

Table 3.1.1. Infrastructure design objectives

Requirement	Description	Technologies / Solutions used
Logical network segmentation	Separating users, servers, management devices, and the DMZ into distinct logical networks	VLANs
Routing between segments	Providing controlled communication between VLANs and different subnets	Layer 3 switches, SVIs
Gateway redundancy	Maintaining a stable gateway for end stations if a Layer 3 device fails	HSRP
Dynamic routing	Automatic route exchange between Layer 3 devices and provision of alternative paths	OSPF
Layer 2 redundancy	Preventing loops and maintaining connectivity through alternative links	Rapid STP, EtherChannel
Link aggregation	Grouping multiple physical connections into a single logical link for availability and performance	EtherChannel, LACP
Access control	Restricting unauthorized traffic between VLANs and functional areas	ACLs
DMZ isolation	Separating the controlled exposed service from internal servers and the Campus network	DMZ VLAN, ACLs
Providing network services	Implementing services used to test infrastructure operation	Web, DNS, FTP, E-mail
Device management	Separating management traffic from standard user traffic	Management VLAN

By meeting these requirements, the designed infrastructure can demonstrate the operation of a segmented, redundant, and secure Data Center-Campus network. The topology supports both controlled access to services and the testing of failover and isolation mechanisms between network areas.

3.2 SELECTION OF THE DATA CENTER-CAMPUS ARCHITECTURE

The Data Center-Campus architecture was selected because it allows the infrastructure to be organized into distinct functional areas according to the roles of users, devices, and services. In an organizational network, user stations, management devices, internal servers, and services exposed in a controlled manner should not be treated as part of the same logical area. Separating them improves manageability, traffic control, and security.

The Campus area is intended to connect users and end devices. It includes workstations and devices used to access network resources. For clearer organization, the Campus area is divided into two layers: Campus Distribution and Campus Access. The Campus Access layer connects end devices, while Campus Distribution aggregates traffic from the access switches and forwards it toward the central part of the infrastructure.

The Core/Distribution area concentrates traffic between the network's main functional areas. Important functions such as inter-VLAN routing, access-policy enforcement, and redundant VLAN gateways are implemented here. A combined Core/Distribution layer is appropriate for a small or medium-sized infrastructure because it reduces topology complexity while preserving the essential principles of hierarchical design.

The Routing Layer provides Layer 3 connectivity and route exchange between the main devices. In this project, the area supports dynamic routing and path redundancy so that the network can maintain communication even when problems occur on selected links.

The Data Center area hosts the network's internal services. It contains servers providing services such as Web, DNS, FTP, and E-mail. Placing the servers in a dedicated area protects important resources and enables clear access policies between users and services.

In addition to the Campus and Data Center areas, the architecture includes a DMZ. This area hosts a service exposed in a controlled manner and kept separate from internal servers. With this organization, services with a higher level of exposure do not provide direct access to the internal infrastructure, and traffic to and from the area can be filtered through security policies.

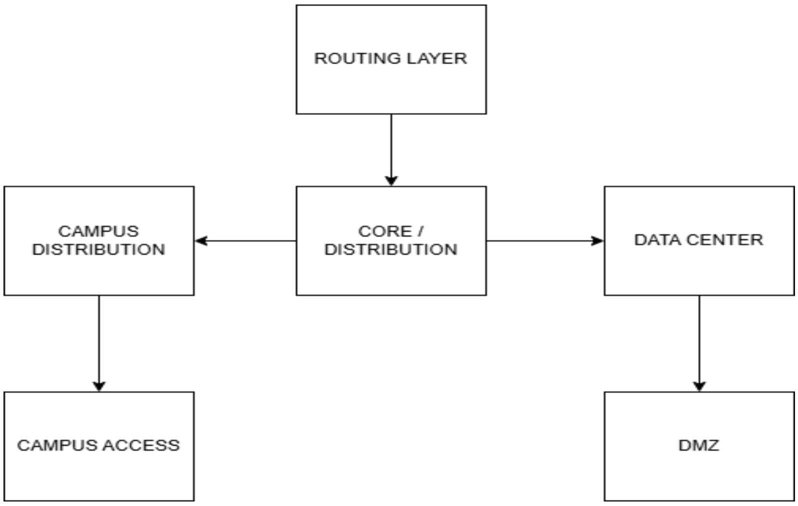

Figure 3.2.1 - Logical diagram of the functional areas in the Data Center-Campus architecture

Figure 3.2.1 shows how the infrastructure is divided into functional areas. The Routing Layer supports Layer 3 routing and connectivity, the Core/Distribution area forms the central point of the infrastructure, and the Campus is divided into Campus Distribution and Campus Access. The Data Center hosts the internal servers, while the DMZ is separated for services that must be accessed in a controlled manner. This organization clearly separates network roles and facilitates the implementation of routing, redundancy, and security mechanisms.

The choice of this architecture is justified by the project objective of building a segmented, redundant, and secure Data Center-Campus infrastructure. The selected model enables an integrated demonstration of VLANs, inter-VLAN routing, HSRP redundancy, dynamic OSPF routing, Layer 2 mechanisms such as Rapid STP and EtherChannel, and access policies implemented through ACLs.

3.3 TOPOLOGY AND DESIGN MODEL

The topology designed in Cisco Packet Tracer represents a Data Center-Campus infrastructure organized into distinct functional areas. The network structure includes end users, distribution and access devices, Layer 3 switches, redundant routers, internal servers, and a separate DMZ.

The design model is hierarchical and modular. The network is divided into the Routing Layer, Core/Distribution Layer, Campus Distribution, Campus Access, Data Center, and DMZ. Each area has a clearly defined role, and this separation enables clearer infrastructure administration and more effective application of routing, redundancy, and security policies.

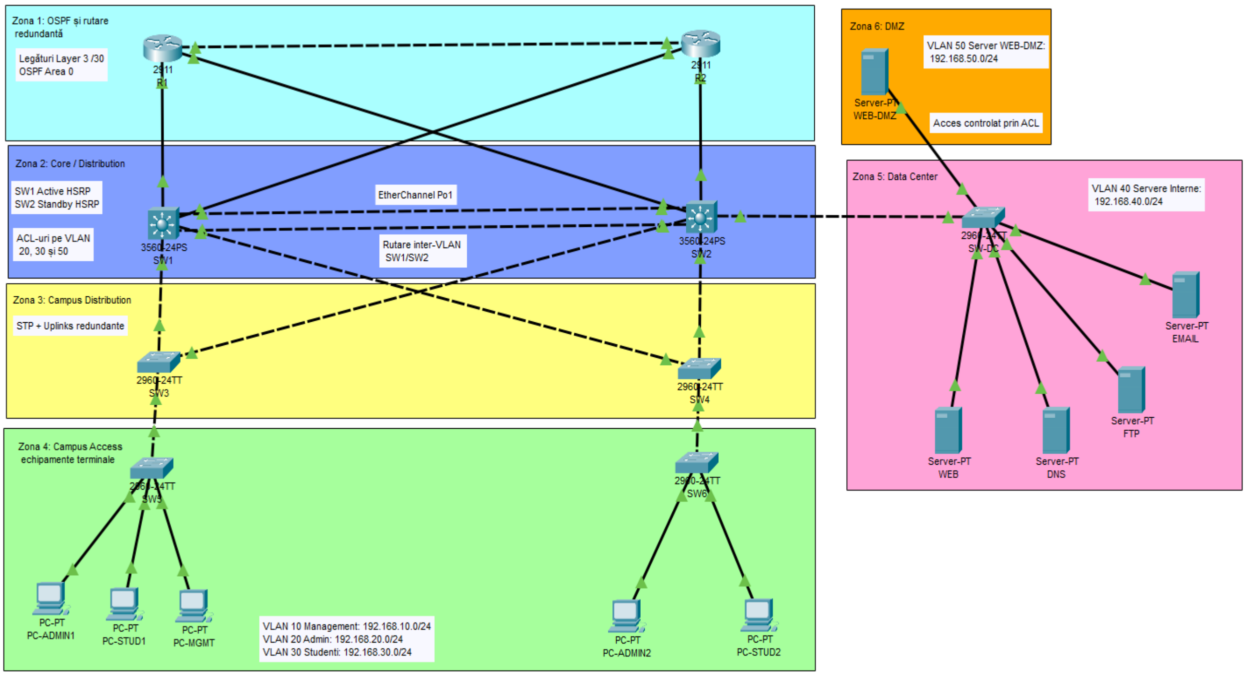

Figure 3.3.1 - Overall topology of the Data Center-Campus network designed in Cisco Packet Tracer

Figure 3.3.1 presents the complete topology of the designed network. The upper part contains the routing area, formed by routers R1 and R2, which are used for dynamic routing with OSPF and to provide redundant paths. Below this area is the Core/Distribution layer, formed by Layer 3 switches SW1 and SW2, which perform inter-VLAN routing, provide redundant gateways through HSRP, and enforce access policies through ACLs.

The Campus area is divided into Campus Distribution and Campus Access. Switches SW3 and SW4 aggregate traffic from the access area, while switches SW5 and SW6 connect user end stations. This area contains devices belonging to the Management, Admin, and Students VLANs, each with different roles and access levels.

The Data Center area is separated from the Campus area and hosts the infrastructure's internal servers. Web, DNS, FTP, and E-mail servers are located here and are used to test network services and connectivity between users and resources. Placing the servers in a dedicated VLAN makes it possible to control traffic to these services.

The DMZ is represented as a separate area and contains the Web-DMZ server. It is isolated from the internal servers and is intended for controlled access. By including this area, the topology distinguishes between internal services located in the Data Center and services exposed in a controlled manner in a separate zone.

At the design level, the topology includes several redundancy mechanisms. Routers R1 and R2 are redundantly connected to the Layer 3 switches, SW1 and SW2 operate as redundant central devices, and multiple links are present between switching devices. OSPF provides route exchange between Layer 3 devices, HSRP provides gateway availability for the VLANs, and Rapid STP and EtherChannel help maintain Layer 2 connectivity.

3.4 DESCRIPTION OF THE FUNCTIONAL AREAS

The designed topology is divided into several functional areas, each with a specific role within the infrastructure. This separation provides clearer device organization, isolates traffic according to the role of each area, and allows routing, redundancy, and security mechanisms to be applied at the appropriate points in the network. The main areas of the topology are the OSPF and redundant routing area, Core / Distribution, Campus Distribution, Campus Access, Data Center, and DMZ.

3.4.1 OSPF AND REDUNDANT ROUTING AREA

The Routing Layer is the upper layer of the topology and provides Layer 3 connectivity between the main network devices. It consists of routers R1 and R2, which are connected to each other and to Layer 3 switches SW1 and SW2.

OSPF is used in this area so that the routers and Layer 3 switches can automatically exchange routing information. As a result, the infrastructure can use multiple available paths between devices without depending on a single link. If a connection fails, OSPF can recalculate the route and maintain communication through an alternative path.

The links between the routers and Layer 3 switches use Layer 3 transit networks for point-to-point communication between routing devices. This organization provides a clear connection structure and supports infrastructure-level redundancy.

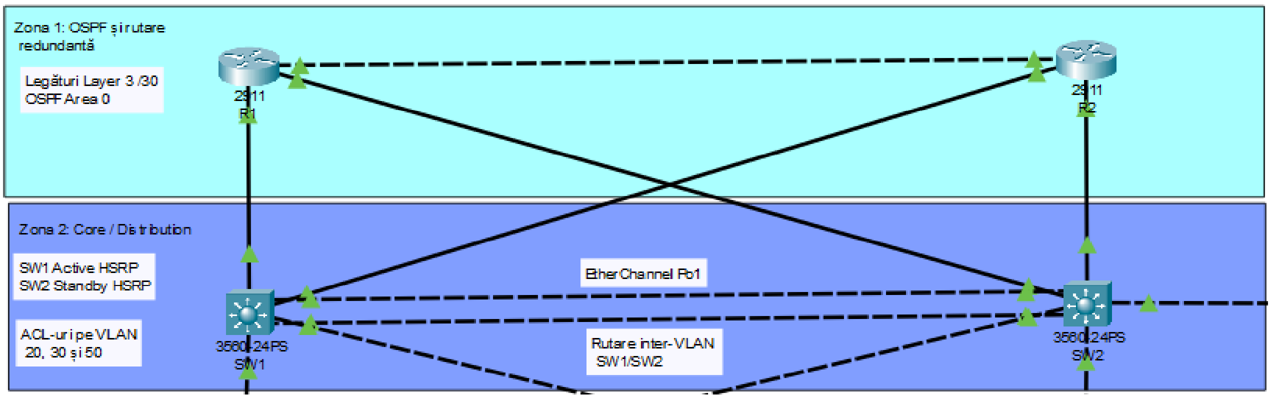

Figure 3.4.1.1 - Routing Layer designed in Cisco Packet Tracer

Figure 3.4.1.1 shows routers R1 and R2 together with their links to Layer 3 switches SW1 and SW2. This area enables OSPF route exchange and provides alternative paths if a failure occurs on one of the links.

3.4.2 CORE / DISTRIBUTION AREA

The Core / Distribution area is the central part of the topology and consists of Layer 3 switches SW1 and SW2. These devices connect the main network areas and provide routing between VLANs.

Essential infrastructure functions are implemented in this area, including inter-VLAN routing, redundant gateway configuration through HSRP, and access-policy enforcement through ACLs. Traffic between users, servers, and the DMZ can therefore be controlled according to the defined rules.

Using two Layer 3 switches increases network availability. If one device becomes unavailable, the other can maintain the virtual gateways and connectivity between the infrastructure's main areas.

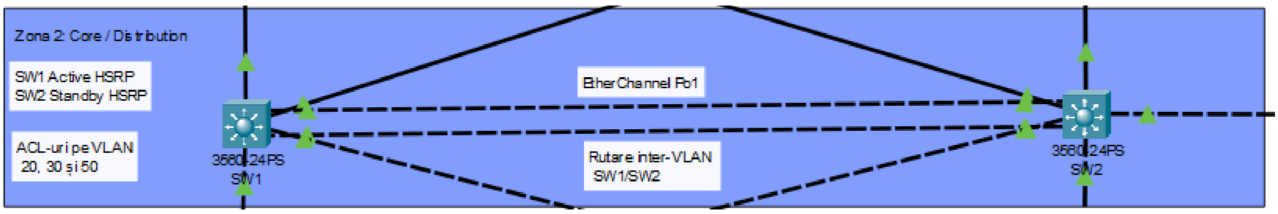

Figure 3.4.2.1 - Core / Distribution area in the designed topology

Figure 3.4.2.1 shows Layer 3 switches SW1 and SW2, which form the central area of the network. These devices perform inter-VLAN routing, participate in HSRP for redundant gateways, and allow ACLs to be applied to control traffic between VLANs, the Data Center, and the DMZ.

3.4.3 CAMPUS DISTRIBUTION AREA

The Campus Distribution area consists of switches SW3 and SW4 and aggregates traffic from the Campus Access area. It connects the access switches, where end stations are attached, to the Core / Distribution area, where routing and traffic control are performed.

By using two distribution switches, the Campus infrastructure does not depend on a single intermediate device. Redundant links toward the central area help maintain connectivity if one connection or one of the switches becomes unavailable.

The Campus Distribution area helps organize user traffic before it reaches the Layer 3 switches. Traffic from the Management, Admin, and Students VLANs can therefore be carried toward the Core / Distribution area, where inter-VLAN routing and access policies are applied.

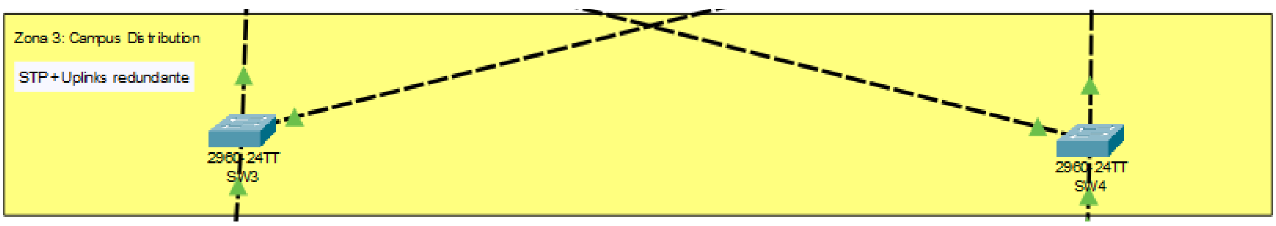

Figure 3.4.3.1 - Campus Distribution area in the designed topology

Figure 3.4.3.1 shows switches SW3 and SW4, which form the distribution layer of the Campus area. These devices aggregate traffic from the access switches and forward it toward the Core / Distribution area. Through its redundant links, this area contributes to infrastructure availability and helps maintain connectivity between users and the rest of the network.

3.4.4 CAMPUS ACCESS AREA

The Campus Access area is the layer at which user end devices are connected. It consists of switches SW5 and SW6, to which stations in the Management, Admin, and Students VLANs are connected.

The main role of this area is to provide users with access to the network infrastructure. By connecting end stations to access switches, users can communicate with internal servers, services available in the Data Center, or other areas permitted by the security policies.

Separating end stations into distinct VLANs allows different access levels to be applied. For example, management devices, administrative users, and users in the Students area are not placed in the same logical segment. This makes traffic easier to organize and allows access to sensitive resources to be controlled through rules defined in the Core / Distribution area.

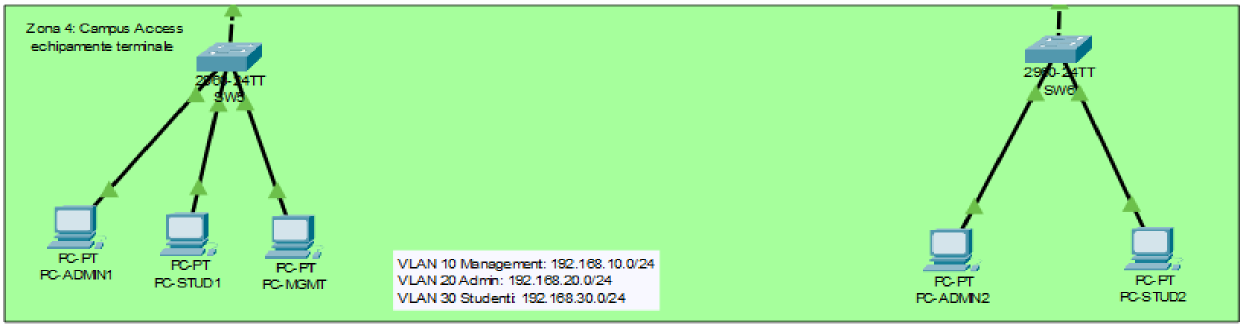

Figure 3.4.4.1 - Campus Access area in the designed topology

Figure 3.4.4.1 shows the Campus network access area, formed by switches SW5 and SW6 and the end stations connected to them. This area connects users to the infrastructure and assigns them to different VLANs according to the role of each device. Traffic from the Campus Access area is forwarded to Campus Distribution and then to the Core / Distribution area, where routing and access policies are applied.

3.4.5 DATA CENTER AREA

The Data Center area hosts the infrastructure's internal servers. In the designed topology, this area includes switch SW-DC and the servers providing network services such as Web, DNS, FTP, and E-mail.

The primary role of the Data Center area is to provide users with internal services that are accessible in a controlled manner from the Campus area. The servers are placed in a dedicated VLAN, separate from user VLANs, which allows clearer traffic administration and network-level access policies.

By separating servers into a dedicated area, the infrastructure can control communication between users and services more effectively. For example, selected user categories can be granted access to Web, DNS, or E-mail services, while FTP access can be restricted according to the defined policies.

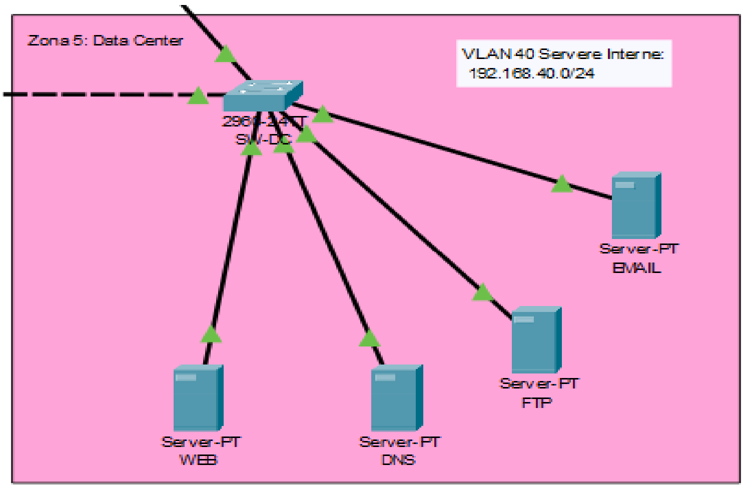

Figure 3.4.5.1 - Data Center area in the designed topology

Figure 3.4.5.1 shows the network's Data Center area, where the internal servers are connected through switch SW-DC. This area hosts the Web, DNS, FTP, and E-mail services used to test infrastructure operation. Placing the servers in a dedicated segment enables access control and protects internal resources from unauthorized traffic.

3.4.6 DMZ AREA

The DMZ is intended for services that must be accessible in a controlled manner but should not be placed in the same area as internal servers. In the designed topology, this area contains the WEB-DMZ server, which is separated from the Data Center area.

The main role of the DMZ is to isolate the exposed service from the rest of the internal infrastructure. Through this separation, the server in the DMZ can be accessed according to the defined policies without providing direct access to critical Data Center servers or management devices.

Traffic to and from the DMZ is controlled through access policies. Only the required communications, such as HTTP access to the WEB-DMZ server, can be permitted, while unauthorized communications toward the internal network can be blocked through ACLs.

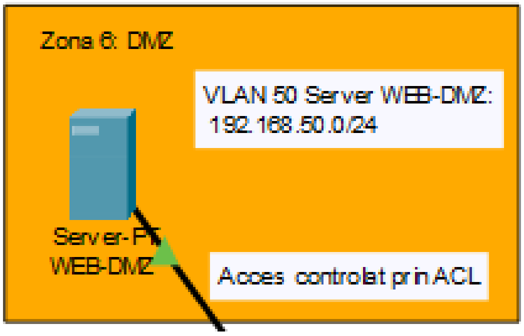

Figure 3.4.6.1 - DMZ area in the designed topology

Figure 3.4.6.1 presents the topology's DMZ area, where the WEB-DMZ server is located. This area is separated from the internal Data Center servers and demonstrates controlled access to an exposed service. By isolating the DMZ and applying access policies, the infrastructure reduces the risk of unauthorized traffic reaching internal network resources directly.

3.5 EQUIPMENT USED AND ITS ROLE

3.5.1 ROUTERS

Routers are network devices that operate at Layer 3 of the OSI model and interconnect different networks. They examine packet IP addresses and determine the path traffic must follow to reach its destination. In a network infrastructure, routers can be used to connect multiple subnets, provide access to other networks, and implement static or dynamic routing [16].

Figure 3.5.1.1 - Cisco 2900 Series router. (source: https://www.cisco.com/c/en/us/support/routers/2900-series-integrated-services-routers-isr/series.html)

In this project, routers R1 and R2 are used in the upper part of the topology, referred to as the OSPF and redundant routing area. They participate in dynamic routing through OSPF and are redundantly connected to Layer 3 switches SW1 and SW2. Their role is to exchange routes between the main devices and provide alternative paths if a link fails.

3.5.2 SWITCHES

Switches are devices used to connect devices within a local network. Switches typically operate at Layer 2 and forward Ethernet frames based on MAC addresses. They connect end stations, servers, and other network devices, helping organize traffic within a LAN. Switches can also support VLANs, allowing devices connected to the same physical infrastructure to be logically separated [17].

Figure 3.5.2.1 - Cisco 3560 Series switch (source: https://www.cisco.com/c/en/us/support/switches/catalyst-3560-series-switches/series.html)

Both Layer 3 and Layer 2 switches are used in the project. Layer 3 switches SW1 and SW2 have a central role in the infrastructure. They perform inter-VLAN routing, provide virtual gateways through HSRP, and allow access policies to be applied through ACLs. Using two Layer 3 switches provides gateway redundancy and connectivity between the main areas of the topology.

Layer 2 switches are used in the Campus and Data Center areas. SW3 and SW4 act as distribution switches by aggregating traffic from the access area, while SW5 and SW6 connect user end stations. Switch SW-DC connects the internal Data Center servers. These devices carry VLAN traffic and connect devices to the network's central infrastructure.

3.5.3 SERVERS

Servers are specialized devices or systems that provide services to other network devices. They can host applications, web pages, e-mail services, DNS services, files, or databases. In an organizational infrastructure, servers are generally placed in a dedicated area so that they can be managed and protected more effectively [3].

In this project, servers provide Web, DNS, FTP, and E-mail services. They are located in the Data Center area and connected through switch SW-DC. The servers make it possible to test infrastructure operation because users in different VLANs can attempt to access the services according to the defined security policies.

Separate from the internal servers, the WEB-DMZ server is placed in the DMZ. It represents a Web service exposed in a controlled manner and is isolated from the internal Data Center servers. This separation makes it possible to test the difference between access to internal services and access to a service placed in a separate area protected by ACL policies.

3.6 VLAN AND SUBNET PLAN

Separate VLANs were defined for logical segmentation of the infrastructure, with each VLAN corresponding to a user category or functional area. This division organizes traffic, isolates important areas, and enables different access policies to be applied between network segments.

IP addressing was selected from the private 192.168.0.0/16 address space, which is commonly used in local networks. These addresses are suitable for an internal infrastructure because they are not routed directly on the Internet. The project uses the 192.168.X.0/24 scheme, where X corresponds to the VLAN number. For example, VLAN 10 uses subnet 192.168.10.0/24, while VLAN 40 uses subnet 192.168.40.0/24.

The /24 mask, equivalent to 255.255.255.0, was selected for simplicity and clarity. It provides up to 254 usable addresses in each VLAN, which is sufficient for the size of the topology designed in Cisco Packet Tracer. Using the same mask for all VLANs also simplifies configuration and testing.

Table 3.6.1 - VLAN and subnet plan

VLAN	Name	Subnet	Role
10	Management	192.168.10.0/24	Network device administration
20	Admin	192.168.20.0/24	Administrative users
30	Students	192.168.30.0/24	Standard users with limited access
40	Internal Servers	192.168.40.0/24	Web, DNS, FTP, and E-mail services
50	DMZ	192.168.50.0/24	Web server exposed in a controlled manner

With this structure, each category of device has its own subnet, making the infrastructure easier to manage and test. Routing between VLANs is performed by the Layer 3 switches, while access between segments is subsequently controlled through ACL policies.

3.7 IP ADDRESSING PLAN

The IP addressing plan defines how VLANs, network devices, servers, and Layer 3 links are identified in the topology. Each functional area in the project uses a separate subnet to provide clear organization and easier administration.

The VLANs use /24 networks, and communication between them is handled by the Layer 3 switches. /30 transit networks are used for point-to-point links between routers and Layer 3 switches because they are suitable for connections between two devices.

Accordingly, 192.168.x.0/24 networks are used for VLANs and servers, while 10.0.x.x/30 networks are used for routing links between Layer 3 devices.

3.7.1 VLAN AND GATEWAY ADDRESSING

A separate /24 subnet was allocated to each VLAN in the topology. The default gateway used by devices in each VLAN is the HSRP virtual address, while Layer 3 switches SW1 and SW2 have their own addresses for each VLAN interface.

Table 3.7.1.1 - VLAN and gateway addressing

VLAN	Name	Subnet	HSRP virtual gateway	SW1 address	SW2 address
10	Management	192.168.10.0/24	192.168.10.1	192.168.10.2	192.168.10.3
20	Admin	192.168.20.0/24	192.168.20.1	192.168.20.2	192.168.20.3
30	Students	192.168.30.0/24	192.168.30.1	192.168.30.2	192.168.30.3
40	Internal Servers	192.168.40.0/24	192.168.40.1	192.168.40.2	192.168.40.3
50	DMZ	192.168.50.0/24	192.168.50.1	192.168.50.2	192.168.50.3

With this addressing scheme, each logical network area is separated into its own subnet, while the virtual gateway provides a common access point for devices in the corresponding VLAN.

3.7.2 LAYER 3 TRANSIT NETWORKS

Layer 3 transit networks are used to directly connect two routing devices, such as two routers or a router and a Layer 3 switch. They are not intended for end users, but for communication between devices participating in the routing process. /30 transit networks were used for the links between routers and Layer 3 switches.

Table 3.7.2.1 - Layer 3 transit networks used in the topology

Link	Transit network	First device address	Second device address
R1 – R2	10.0.0.0/30	R1, G0/0 – 10.0.0.1	R2, G0/0 – 10.0.0.2

Link	Transit network	First device address	Second device address
R1 – SW1	10.0.1.0/30	R1, G0/1 – 10.0.1.1	SW1, G0/1 – 10.0.1.2
R1 – SW2	10.0.1.4/30	R1, G0/2 – 10.0.1.5	SW2, G0/1 – 10.0.1.6
R2 – SW1	10.0.1.8/30	R2, G0/1 – 10.0.1.9	SW1, G0/2 – 10.0.1.10
R2 – SW2	10.0.1.12/30	R2, G0/2 – 10.0.1.13	SW2, G0/2 – 10.0.1.14

Through these transit networks, Layer 3 devices can communicate with one another and exchange routes through OSPF. The presence of multiple links allows connectivity to be maintained even if one connection becomes unavailable.

3.7.3 ADDRESSING OF NETWORK DEVICES, SERVERS, AND END STATIONS

In addition to VLAN and transit-network addresses, the topology includes static addresses for network devices, servers, and the end stations used for testing. These addresses identify each device and enable connectivity between the functional areas of the infrastructure to be verified.

Table 3.7.3.1 - Network device addressing

Device	Functional area	Management IP address	Gateway
SW3	Campus Distribution	192.168.10.11/24	192.168.10.1
SW4	Campus Distribution	192.168.10.12/24	192.168.10.1
SW5	Campus Access	192.168.10.21/24	192.168.10.1
SW6	Campus Access	192.168.10.22/24	192.168.10.1
SW-DC	Data Center	192.168.10.30/24	192.168.10.1

Table 3.7.3.2 - Server and end-station addressing

Device	VLAN	IP address	Gateway	DNS
WEB	VLAN 40 - Servers	192.168.40.10/24	192.168.40.1	192.168.40.11
DNS	VLAN 40 - Servers	192.168.40.11/24	192.168.40.1	-
FTP	VLAN 40 - Servers	192.168.40.12/24	192.168.40.1	192.168.40.11
EMAIL	VLAN 40 - Servers	192.168.40.13/24	192.168.40.1	192.168.40.11
WEB-DMZ	VLAN 50 - DMZ	192.168.50.10/24	192.168.50.1	192.168.40.11
PC-MGMT	VLAN 10 - Management	192.168.10.100/24	192.168.10.1	192.168.40.11
PC-ADMIN1	VLAN 20 - Admin	192.168.20.10/24	192.168.20.1	192.168.40.11
PC-ADMIN2	VLAN 20 - Admin	192.168.20.11/24	192.168.20.1	192.168.40.11

Device	VLAN	IP address	Gateway	DNS
PC-STUD1	VLAN 30 - Students	192.168.30.10/24	192.168.30.1	192.168.40.11
PC-STUD2	VLAN 30 - Students	192.168.30.11/24	192.168.30.1	192.168.40.11

With this addressing scheme, every important device in the topology can be clearly identified. Servers are grouped in the Data Center area, the WEB-DMZ server is separated into the DMZ, and end stations are placed in different VLANs to test access policies.

3.8 SECURITY AND ISOLATION POLICIES

Security policies were defined according to the role of each area in the topology. Their purpose is to permit access only to required services and block unauthorized traffic toward sensitive parts of the infrastructure.

Access control is implemented on the Layer 3 switches through rules applied to VLAN interfaces. Traffic between VLANs is therefore not freely permitted; instead, it is filtered according to user category and the services those users need to access.

The Management VLAN is protected from standard users because it is intended for administration of network devices. Unauthorized access to this area is restricted to reduce the risk of configuration changes or access to critical devices.

More restrictive rules were applied to users in the Students VLAN. They can access only the required services, such as DNS, the internal Web server, and E-mail, but they cannot access the management area, the FTP server, or the DMZ. This separation limits the ability of a standard user to reach resources that are not required for their role.

Users in the Admin VLAN have a broader level of access to internal servers because this segment represents an administrative area of the network. However, access to the DMZ remains controlled in order to preserve separation between the internal area and services exposed in a controlled manner.

The Data Center is treated as an important internal area because it hosts the infrastructure's main servers. In the project, this area corresponds to VLAN 40 and contains the Web, DNS, FTP, and E-mail servers. Each service has a clear role in testing network operation: the Web server is used for HTTP access, the DNS server for domain-name resolution, the FTP server for file transfers, and the E-mail server for testing e-mail communication.

Access to the Data Center is not identical for all VLANs. Users in the Admin area have broader access to internal servers because they represent an administrative network segment. By contrast, users in the Students area have access only to required services such as DNS, the internal Web server, and E-mail, while FTP access is restricted. This approach allows internal servers to be used by the user categories that need them without providing general access to every resource.

The DMZ is separate from the Data Center and corresponds to VLAN 50. It contains the WEB-DMZ server, which represents a Web service exposed in a controlled manner. Separating this server from the internal servers creates a clear boundary between the organization's internal services and services that may have a higher level of exposure.

Under the access policies, traffic to the WEB-DMZ server is permitted only under controlled conditions. Administrative users can access the Web service in the DMZ through HTTP/HTTPS, while users in the Students area cannot access this zone. Traffic initiated from the DMZ toward internal networks is also restricted, except for communications required by the configured services, such as DNS resolution or return traffic for permitted connections. The DMZ therefore remains isolated from the internal infrastructure and reduces the risk that an exposed service could provide direct access to internal servers or management devices.

Through these policies, the designed infrastructure separates traffic between users, servers, management devices, and the DMZ. The exact access rules and their validation are presented in the chapters covering network implementation and testing.

3.9 NETWORK REDUNDANCY STRATEGY

The network redundancy strategy was designed to reduce the infrastructure's dependence on a single device or link. The topology therefore includes redundancy mechanisms at both Layer 3 and Layer 2.

At Layer 3, redundancy is implemented using routers R1 and R2, which are connected to Layer 3 switches SW1 and SW2. OSPF automatically exchanges routes between these devices and allows alternative paths to be used if a link fails.

In the Core / Distribution area, switches SW1 and SW2 provide redundant gateways for the VLANs through HSRP. End stations therefore use the same gateway address even though the active role can be assumed by either of the two Layer 3 switches.

At Layer 2, redundancy is supported by multiple links between switches. Rapid STP prevents loops, while EtherChannel groups multiple physical connections into a single logical link, increasing the availability of the connection between devices.

By combining OSPF, HSRP, Rapid STP, and EtherChannel, the designed infrastructure can maintain connectivity if a link or an important device fails. The configuration and testing of these mechanisms are presented in the implementation chapter.

4. NETWORK IMPLEMENTATION AND CONFIGURATION

4.1 CISCO PACKET TRACER SIMULATION ENVIRONMENT

The network was implemented in Cisco Packet Tracer, a simulation environment used to design and test network infrastructures. It allows topologies containing routers, switches, servers, and end stations to be built without requiring physical equipment.

Cisco Packet Tracer is used primarily in educational environments and is associated with the Cisco Networking Academy program. Its main purpose is to help users understand network operation by simulating practical scenarios. It can be used to configure devices, test connections, and observe packet paths within a virtual network.

Packet Tracer was selected for this thesis because it provides the functionality required to implement a Data Center-Campus infrastructure. Technologies such as VLANs, inter-VLAN routing, OSPF, HSRP, STP, EtherChannel, and ACLs were used in the project and can all be configured and tested directly in the simulator.

Another reason for selecting this environment is that it allows configurations to be verified quickly. Commands such as show ip interface brief, show vlan brief, show ip route, and show access-lists make it possible to inspect device status and identify potential configuration errors. Cisco Packet Tracer therefore provides a suitable environment for network design, configuration, and validation.

4.2 INITIAL CONFIGURATION OF LAYER 3 DEVICES

The initial configuration of the Layer 3 devices included assigning hostnames, enabling interfaces, and configuring IP addresses for the transit links. These settings provide the foundation for dynamic routing and for connectivity between routers and Layer 3 switches.

On routers R1 and R2, the GigabitEthernet interfaces used for connections to the other Layer 3 devices were configured.

```
R1>
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#
R1(config)#interface GigabitEthernet0/0
R1(config-if)#description LINK_TO_R2
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet0/1
R1(config-if)#description LINK_TO_SW1
R1(config-if)#ip address 10.0.1.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet0/2
R1(config-if)#description LINK_TO_SW2
R1(config-if)#ip address 10.0.1.5 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     10.0.1.1        YES manual up                    up
GigabitEthernet0/2     10.0.1.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Figure 4.2.1 - Layer 3 interface configuration on router R1

Figure 4.2.1 shows the configuration of router R1's interfaces and verification of their status using the show ip interface brief command. The configured interfaces are active and use the IP addresses defined in the addressing plan.

```
R2>
R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#
R2(config)#interface GigabitEthernet0/0
R2(config-if)#description LINK_TO_R1
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface GigabitEthernet0/1
R2(config-if)#description LINK_TO_SW1
R2(config-if)#ip address 10.0.1.9 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface GigabitEthernet0/2
R2(config-if)#description LINK_TO_SW2
R2(config-if)#ip address 10.0.1.13 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#end
R2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R2#show ip in brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.2        YES manual up                    up
GigabitEthernet0/1     10.0.1.9        YES manual up                    up
GigabitEthernet0/2     10.0.1.13       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Figure 4.2.2 - Layer 3 interface configuration on router R2.

Figure 4.2.2 highlights the configuration of router R2's interfaces. R2 is connected to R1, SW1, and SW2, providing redundant links in the routing area.

IP routing was enabled on Layer 3 switches SW1 and SW2, and the interfaces connected to the routers were configured as Layer 3 interfaces using the no switchport command.

IOS Command Line Interface

```
SW1>
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#hostname SW1
SW1(config)#
SW1(config)#ip routing
SW1(config)#
SW1(config)#interface GigabitEthernet0/1
SW1(config-if)#description LINK_TO_R1
SW1(config-if)#no switchport
SW1(config-if)#ip address 10.0.1.2 255.255.255.252
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface GigabitEthernet0/2
SW1(config-if)#description LINK_TO_R2
SW1(config-if)#no switchport
SW1(config-if)#ip address 10.0.1.10 255.255.255.252
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

Figure 4.2.3 - Layer 3 interface configuration on switch SW1.

Figure 4.2.3 shows IP routing being enabled on SW1 and the configuration of the interfaces used for the connections to routers R1 and R2.

```
SW2>
SW2>
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#hostname SW2
SW2(config)#
SW2(config)#ip routing
SW2(config)#
SW2(config)#interface GigabitEthernet0/1
SW2(config-if)#description LINK_TO_R1
SW2(config-if)#no switchport
SW2(config-if)#ip address 10.0.1.6 255.255.255.252
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface GigabitEthernet0/2
SW2(config-if)#description LINK_TO_R2
SW2(config-if)#no switchport
SW2(config-if)#ip address 10.0.1.14 255.255.255.252
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#end
SW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
SW2#
```

Figure 4.2.4 - Layer 3 interface configuration on switch SW2.

Figure 4.2.4 shows the Layer 3 interface configuration on SW2. With these settings, SW2 can participate in Layer 3 communication together with the routers and switch SW1.

4.3 VLAN AND TRUNK LINK CONFIGURATION

At this stage, the VLANs used to separate the functional areas of the network and the trunk links required to carry them between switches were configured. The configuration was implemented according to the role of each switch in the topology.

On switch SW1, the main infrastructure VLANs were created: Management, Admin, Students, Servers, and DMZ. In addition, the ports toward the Campus Distribution area and the logical Port-channel link toward SW2 were configured as trunks.

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#vlan 10
SW1(config-vlan)#name MANAGEMENT
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name ADMIN
SW1(config-vlan)#exit
SW1(config)#vlan 30
SW1(config-vlan)#name STUDENTI
SW1(config-vlan)#exit
SW1(config)#vlan 40
SW1(config-vlan)#name SERVERS
SW1(config-vlan)#exit
SW1(config)#vlan 50
SW1(config-vlan)#name DMZ
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#interface FastEthernet0/1
SW1(config-if)#description TRUNK_TO_SW3
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface FastEthernet0/2
SW1(config-if)#description TRUNK_TO_SW4
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Port-channel1
SW1(config-if)#description ETHERCHANNEL_TO_SW2
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
```

Figure 4.3.1 - VLAN and trunk link configuration on SW1

Figure 4.3.1 shows the commands used to create the VLANs and configure the trunk ports on SW1. These links carry the VLANs toward the other network areas.

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
10   MANAGEMENT                       active
20   ADMIN                            active
30   STUDENTI                         active
40   SERVERS                          active
50   DMZ                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#show interfaces trunk
%SYS-5-CONFIG_I: Configured from console by console

Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

Port        Vlans allowed and active in management domain
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

SW1#
```

Figure 4.3.2 - Verification of VLANs and trunks on SW1

Figure 4.3.2 confirms the presence of the configured VLANs and the active state of SW1's trunk links.

On switch SW4, located in the Campus Distribution area, the required VLANs and the trunk links toward the Core / Distribution area and the Campus Access area were configured.

```
SW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW4(config)#
SW4(config)#vlan 10
SW4(config-vlan)#name MANAGEMENT
SW4(config-vlan)#exit
SW4(config)#vlan 20
SW4(config-vlan)#name ADMIN
SW4(config-vlan)#exit
SW4(config)#vlan 30
SW4(config-vlan)#name STUDENTI
SW4(config-vlan)#exit
SW4(config)#vlan 40
SW4(config-vlan)#name SERVERS
SW4(config-vlan)#exit
SW4(config)#vlan 50
SW4(config-vlan)#name DMZ
SW4(config-vlan)#exit
SW4(config)#
SW4(config)#interface FastEthernet0/1
SW4(config-if)#description TRUNK_TO_CORE
SW4(config-if)#switchport mode trunk
SW4(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW4(config-if)#no shutdown
SW4(config-if)#exit
SW4(config)#
SW4(config)#interface FastEthernet0/2
SW4(config-if)#description TRUNK_TO_CORE
SW4(config-if)#switchport mode trunk
SW4(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW4(config-if)#no shutdown
SW4(config-if)#exit
SW4(config)#
SW4(config)#interface FastEthernet0/3
SW4(config-if)#description TRUNK_TO_ACCESS
SW4(config-if)#switchport mode trunk
SW4(config-if)#switchport trunk allowed vlan 10,20,30
SW4(config-if)#no shutdown
SW4(config-if)#exit
SW4(config)#
SW4(config)#end
SW4#write memory
```

Figure 4.3.3 - VLAN and trunk link configuration on SW4

Figure 4.3.3 shows the trunk configuration on SW4. The ports toward the central area carry VLANs 10, 20, 30, 40, and 50, while the link toward the access area carries VLANs 10, 20, and 30.

On switch SW5, located in the Campus Access area, the VLANs for the end stations and their corresponding access ports were configured. The uplink toward the distribution layer was configured as a trunk.

```
SW5>enable
SW5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW5(config)#
SW5(config)#vlan 10
SW5(config-vlan)#name MANAGEMENT
SW5(config-vlan)#exit
SW5(config)#vlan 20
SW5(config-vlan)#name ADMIN
SW5(config-vlan)#exit
SW5(config)#vlan 30
SW5(config-vlan)#name STUDENTI
SW5(config-vlan)#exit
SW5(config)#
SW5(config)#interface FastEthernet0/1
SW5(config-if)#description TRUNK_TO_DISTRIBUTION
SW5(config-if)#switchport mode trunk
SW5(config-if)#switchport trunk allowed vlan 10,20,30
SW5(config-if)#no shutdown
SW5(config-if)#exit
SW5(config)#
SW5(config)#interface FastEthernet0/10
SW5(config-if)#description PC_ADMIN
SW5(config-if)#switchport mode access
SW5(config-if)#switchport access vlan 20
SW5(config-if)#no shutdown
SW5(config-if)#exit
SW5(config)#
SW5(config)#interface FastEthernet0/11
SW5(config-if)#description PC_STUDENT
SW5(config-if)#switchport mode access
SW5(config-if)#switchport access vlan 30
SW5(config-if)#no shutdown
SW5(config-if)#exit
SW5(config)#
SW5(config)#interface FastEthernet0/12
SW5(config-if)#description PC_MANAGEMENT
SW5(config-if)#switchport mode access
SW5(config-if)#switchport access vlan 10
SW5(config-if)#no shutdown
SW5(config-if)#exit
SW5(config)#
SW5(config)#end
SW5#write memory
```

Figure 4.3.4 - Access and trunk port configuration on SW5.

Figure 4.3.4 shows how the access ports are assigned to the VLANs used by the end stations. The ports for Admin, Students, and Management are therefore placed in their corresponding VLANs.

On switch SW-DC, the VLANs for the Data Center and DMZ areas were configured. The internal servers were assigned to VLAN 40, while the WEB-DMZ server was assigned to VLAN 50.

```
SW-DC>enable
SW-DC#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW-DC(config)#
SW-DC(config)#vlan 10
SW-DC(config-vlan)#name MANAGEMENT
SW-DC(config-vlan)#exit
SW-DC(config)#vlan 40
SW-DC(config-vlan)#name SERVERS
SW-DC(config-vlan)#exit
SW-DC(config)#vlan 50
SW-DC(config-vlan)#name DMZ
SW-DC(config-vlan)#exit
SW-DC(config)#
SW-DC(config)#interface FastEthernet0/1
SW-DC(config-if)#description TRUNK_TO_CORE
SW-DC(config-if)#switchport mode trunk
SW-DC(config-if)#switchport trunk allowed vlan 10,40,50
SW-DC(config-if)#no shutdown
SW-DC(config-if)#exit
SW-DC(config)#
SW-DC(config)#interface range FastEthernet0/10 - 13
SW-DC(config-if-range)#description SERVERS_DATA_CENTER
SW-DC(config-if-range)#switchport mode access
SW-DC(config-if-range)#switchport access vlan 40
SW-DC(config-if-range)#no shutdown
SW-DC(config-if-range)#exit
SW-DC(config)#
SW-DC(config)#interface FastEthernet0/14
SW-DC(config-if)#description WEB_DMZ
SW-DC(config-if)#switchport mode access
SW-DC(config-if)#switchport access vlan 50
SW-DC(config-if)#no shutdown
SW-DC(config-if)#exit
SW-DC(config)#
SW-DC(config)#end
SW-DC#write memory
```

Figure 4.3.5 - VLAN configuration for the Data Center and DMZ areas on SW-DC

Figure 4.3.5 shows the port configuration for the internal servers and the WEB-DMZ server. The link toward the Core / Distribution area is configured as a trunk to carry the required VLANs.

Similar configurations were applied to the other switches in the topology according to the VLANs they carry and the role of each device. This stage prepared the network for inter-VLAN routing and for the implementation of redundancy and security mechanisms.

4.4 VLAN INTERFACE AND INTER-VLAN ROUTING CONFIGURATION

After the VLANs and trunk links were configured, VLAN interfaces were configured on Layer 3 switches SW1 and SW2. These interfaces enable routing between VLANs and connect each logical segment to the Core / Distribution area.

On SW1, VLAN interfaces 10, 20, 30, 40, and 50 were configured, each receiving its own IP address according to the addressing plan.

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#ip routing
SW1(config)#
SW1(config)#interface Vlan10
SW1(config-if)#description SVI_MANAGEMENT
SW1(config-if)#ip address 192.168.10.2 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan20
SW1(config-if)#description SVI_ADMIN
SW1(config-if)#ip address 192.168.20.2 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan30
SW1(config-if)#description SVI_STUDENTI
SW1(config-if)#ip address 192.168.30.2 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan40
SW1(config-if)#description SVI_SERVERS
SW1(config-if)#ip address 192.168.40.2 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan50
SW1(config-if)#description SVI_DMZ
SW1(config-if)#ip address 192.168.50.2 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
```

Figure 4.4.1 - VLAN interface configuration on SW1

```
GigabitEthernet0/1     10.0.1.2        YES manual up                    up
GigabitEthernet0/2     10.0.1.10       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.2    YES manual up                    up
Vlan20                 192.168.20.2    YES manual up                    up
Vlan30                 192.168.30.2    YES manual up                    up
Vlan40                 192.168.40.2    YES manual up                    up
Vlan50                 192.168.50.2    YES manual up                    up
SW1#
```

Figure 4.4.2 - Verification of VLAN interfaces on SW1

A similar configuration was implemented on SW2, using different IP addresses for each VLAN interface. SW2 can therefore operate together with SW1 as part of the redundant infrastructure.

```
GigabitEthernet0/1     10.0.1.6        YES manual up                    up
GigabitEthernet0/2     10.0.1.14       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.3    YES manual up                    up
Vlan20                 192.168.20.3    YES manual up                    up
Vlan30                 192.168.30.3    YES manual up                    up
Vlan40                 192.168.40.3    YES manual up                    up
Vlan50                 192.168.50.3    YES manual up                    up
SW2#
```

Figure 4.4.3 - Verification of VLAN interfaces on SW2

```
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50

SW1#
SW1#
```

Figure 4.4.4 - Verification of inter-VLAN routing on SW1

By configuring the VLAN interfaces and enabling ip routing, SW1 and SW2 can perform inter-VLAN routing. The VLAN networks appear as directly connected

networks in the routing table, allowing traffic to be forwarded between the infrastructure's logical segments according to the configured access policies.

4.5 HSRP CONFIGURATION FOR REDUNDANT GATEWAYS

After the VLAN interfaces were configured, HSRP was configured on Layer 3 switches SW1 and SW2. HSRP allows a shared virtual address to be used as the default gateway for each VLAN.

In the project, the HSRP virtual address is the first usable address in each subnet, in the form 192.168.x.1. End stations use this address as their gateway, while SW1 and SW2 have their own addresses in each VLAN.

SW1 was configured with a higher priority so that it operates as the active device for the virtual gateways. SW2 serves as the backup and can assume the gateway role if SW1 becomes unavailable.

```
SW1#
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#interface Vlan10
SW1(config-if)#standby 10 ip 192.168.10.1
SW1(config-if)#standby 10 priority 110
SW1(config-if)#standby 10 preempt
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan20
SW1(config-if)#standby 20 ip 192.168.20.1
SW1(config-if)#standby 20 priority 110
SW1(config-if)#standby 20 preempt
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan30
SW1(config-if)#standby 30 ip 192.168.30.1
SW1(config-if)#standby 30 priority 110
SW1(config-if)#standby 30 preempt
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan40
SW1(config-if)#standby 40 ip 192.168.40.1
SW1(config-if)#standby 40 priority 110
SW1(config-if)#standby 40 preempt
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface Vlan50
SW1(config-if)#standby 50 ip 192.168.50.1
SW1(config-if)#standby 50 priority 110
SW1(config-if)#standby 50 preempt
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
```

Figure 4.5.1 - HSRP configuration on switch SW1

Figure 4.5.1 shows the HSRP virtual-address configuration on SW1 for VLANs 10, 20, 30, 40, and 50. The priority configured on SW1 makes it the active device for these gateways.

```
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#
SW2(config)#interface Vlan10
SW2(config-if)#standby 10 ip 192.168.10.1
SW2(config-if)#standby 10 preempt
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface Vlan20
SW2(config-if)#standby 20 ip 192.168.20.1
SW2(config-if)#standby 20 preempt
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface Vlan30
SW2(config-if)#standby 30 ip 192.168.30.1
SW2(config-if)#standby 30 preempt
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface Vlan40
SW2(config-if)#standby 40 ip 192.168.40.1
SW2(config-if)#standby 40 preempt
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface Vlan50
SW2(config-if)#standby 50 ip 192.168.50.1
SW2(config-if)#standby 50 preempt
SW2(config-if)#exit
SW2(config)#
SW2(config)#end
SW2#write memory
Building configuration...
[OK]
```

Figure 4.5.2 - HSRP configuration on switch SW2

```
SW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           unknown         192.168.10.1
Vl20        20   110 P Active   local           unknown         192.168.20.1
Vl30        30   110 P Active   local           unknown         192.168.30.1
Vl40        40   110 P Active   local           unknown         192.168.40.1
Vl50        50   110 P Active   local           192.168.50.3    192.168.50.1
SW1#
SW1#
```

Figure 4.5.3 - Verification of HSRP status on the Layer 3 switches

Figure 4.5.3 shows the status of the HSRP virtual gateways. The verification makes it possible to observe the active and standby roles for each configured VLAN.

With HSRP configured, the gateway used by end stations remains the same even if the active role is taken over by another Layer 3 switch. This increases network availability and reduces the risk of communication interruption between VLANs.

4.6 OSPF CONFIGURATION

OSPF was configured to provide dynamic routing between the Layer 3 devices. It enables automatic route exchange between routers R1 and R2 and Layer 3 switches SW1 and SW2. In the project, routers R1 and R2 form the routing area, while SW1 and SW2 participate in OSPF through the interfaces connected to the routers and through the configured internal networks.

OSPF was used so that the Layer 3 devices could automatically exchange routing information. The VLAN networks and transit networks can therefore be learned by the main devices without manually configuring every static route.

On router R1, OSPF process 1 was configured with router ID 1.1.1.1. The transit networks connecting R1 to R2, SW1, and SW2 were included in the OSPF process.

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1(config-router)#network 10.0.1.0 0.0.0.3 area 0
R1(config-router)#network 10.0.1.4 0.0.0.3 area 0
R1(config-router)#exit
R1(config)#
R1(config)#end
R1#write memory
Building configuration...
[OK]
R1#
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:38    10.0.0.2        GigabitEthernet0/0
11.11.11.11       1   FULL/DR         00:00:39    10.0.1.2        GigabitEthernet0/1
22.22.22.22       1   FULL/DR         00:00:38    10.0.1.6        GigabitEthernet0/2
R1#show ip route ospf
%SYS-5-CONFIG_I: Configured from console by console

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.1.8 [110/2] via 10.0.0.2, 01:14:51, GigabitEthernet0/0
                 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
O       10.0.1.12 [110/2] via 10.0.0.2, 01:14:51, GigabitEthernet0/0
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
O    192.168.10.0 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
O    192.168.20.0 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
O    192.168.30.0 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
O    192.168.40.0 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
O    192.168.50.0 [110/2] via 10.0.1.2, 01:14:51, GigabitEthernet0/1
                  [110/2] via 10.0.1.6, 01:14:51, GigabitEthernet0/2
```

Figure 4.6.1 - OSPF configuration and verification on router R1

Figure 4.6.1 shows the OSPF configuration on R1 and the OSPF neighbors formed with R2, SW1, and SW2. The routing table also confirms that routes to the internal networks are learned through OSPF.

The same OSPF process was configured on router R2, using router ID 2.2.2.2. It includes the transit networks toward R1, SW1, and SW2.

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2(config-router)#network 10.0.1.8 0.0.0.3 area 0
R2(config-router)#network 10.0.1.12 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#
R2(config)#end
R2#write memory
Building configuration...
[OK]
R2#
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:36    10.0.0.1        GigabitEthernet0/0
11.11.11.11       1   FULL/DR         00:00:36    10.0.1.10       GigabitEthernet0/1
22.22.22.22       1   FULL/DR         00:00:36    10.0.1.14       GigabitEthernet0/2
R2#show ip route ospf
%SYS-5-CONFIG_I: Configured from console by console

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.0.1.0 [110/2] via 10.0.0.1, 01:15:38, GigabitEthernet0/0
                 [110/2] via 10.0.1.10, 01:15:38, GigabitEthernet0/1
O       10.0.1.4 [110/2] via 10.0.0.1, 01:15:48, GigabitEthernet0/0
                 [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
O    192.168.10.0 [110/2] via 10.0.1.10, 01:15:48, GigabitEthernet0/1
                  [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
O    192.168.20.0 [110/2] via 10.0.1.10, 01:15:48, GigabitEthernet0/1
                  [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
O    192.168.30.0 [110/2] via 10.0.1.10, 01:15:48, GigabitEthernet0/1
                  [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
O    192.168.40.0 [110/2] via 10.0.1.10, 01:15:48, GigabitEthernet0/1
                  [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
O    192.168.50.0 [110/2] via 10.0.1.10, 01:15:48, GigabitEthernet0/1
                  [110/2] via 10.0.1.14, 01:15:48, GigabitEthernet0/2
```

Figure 4.6.2 - OSPF configuration and verification on router R2

Figure 4.6.2 highlights the OSPF adjacencies formed by R2 and the learned OSPF routes. The presence of neighbors in the Full state confirms correct operation of the dynamic routing process.

On Layer 3 switch SW1, OSPF was configured with router ID 11.11.11.11. The process includes the transit links toward the routers and the internal VLAN networks that must be advertised through OSPF.

```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#ip routing
SW1(config)#
SW1(config)#router ospf 1
SW1(config-router)#router-id 11.11.11.11
SW1(config-router)#network 10.0.1.0 0.0.0.3 area 0
SW1(config-router)#network 10.0.1.8 0.0.0.3 area 0
SW1(config-router)#network 192.168.10.0 0.0.0.255 area 0
SW1(config-router)#network 192.168.20.0 0.0.0.255 area 0
SW1(config-router)#network 192.168.30.0 0.0.0.255 area 0
SW1(config-router)#network 192.168.40.0 0.0.0.255 area 0
SW1(config-router)#network 192.168.50.0 0.0.0.255 area 0
SW1(config-router)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
SW1#
SW1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:38    10.0.1.1        GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:38    10.0.1.9        GigabitEthernet0/2
SW1#show ip route ospf
%SYS-5-CONFIG_I: Configured from console by console

     10.0.0.0/30 is subnetted, 5 subnets
O       10.0.0.0 [110/2] via 10.0.1.1, 01:17:37, GigabitEthernet0/1
                 [110/2] via 10.0.1.9, 01:17:37, GigabitEthernet0/2
O       10.0.1.4 [110/2] via 10.0.1.1, 01:17:37, GigabitEthernet0/1
O       10.0.1.12 [110/2] via 10.0.1.9, 01:17:37, GigabitEthernet0/2
```

Figure 4.6.3 - OSPF configuration and verification on switch SW1

Figure 4.6.3 shows SW1 participating in the OSPF process. SW1 forms OSPF adjacencies with routers R1 and R2 and contributes to route exchange between the Core / Distribution area and the routing area.

A similar configuration was implemented on Layer 3 switch SW2 using router ID 22.22.22.22. It advertises the same internal VLAN networks and participates in dynamic routing through its links to R1 and R2.

```
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#
SW2(config)#ip routing
SW2(config)#
SW2(config)#router ospf 1
SW2(config-router)#router-id 22.22.22.22
SW2(config-router)#network 10.0.1.4 0.0.0.3 area 0
SW2(config-router)#network 10.0.1.12 0.0.0.3 area 0
SW2(config-router)#network 192.168.10.0 0.0.0.255 area 0
SW2(config-router)#network 192.168.20.0 0.0.0.255 area 0
SW2(config-router)#network 192.168.30.0 0.0.0.255 area 0
SW2(config-router)#network 192.168.40.0 0.0.0.255 area 0
SW2(config-router)#network 192.168.50.0 0.0.0.255 area 0
SW2(config-router)#exit
SW2(config)#
SW2(config)#end
SW2#write memory
Building configuration...
[OK]
SW2#
SW2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:39    10.0.1.5        GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:39    10.0.1.13       GigabitEthernet0/2
SW2#show ip route ospf
%SYS-5-CONFIG_I: Configured from console by console

     10.0.0.0/30 is subnetted, 5 subnets
O       10.0.0.0 [110/2] via 10.0.1.5, 01:18:01, GigabitEthernet0/1
                 [110/2] via 10.0.1.13, 01:18:01, GigabitEthernet0/2
O       10.0.1.0 [110/2] via 10.0.1.5, 01:18:01, GigabitEthernet0/1
O       10.0.1.8 [110/2] via 10.0.1.13, 01:18:01, GigabitEthernet0/2
```

Figure 4.6.4 - OSPF configuration and verification on switch SW2

Figure 4.6.4 confirms the formation of OSPF adjacencies on SW2 and the presence of OSPF routes in the routing table. SW2 can therefore operate together with SW1 in the redundant infrastructure.

With OSPF configured, the Layer 3 devices can automatically exchange routing information. This enables alternative paths to exist between the routing area and the Core / Distribution area, contributing to network redundancy and availability.

4.7 LAYER 2 MECHANISM CONFIGURATION: RAPID STP AND ETHERCHANNEL

Rapid STP and EtherChannel were configured to provide Layer 2 redundancy. Rapid STP was used to prevent loops in the topology, while EtherChannel was used to aggregate the two physical links between SW1 and SW2 into a single logical link.

Rapid-PVST mode was enabled on SW1, and SW1 was configured as the primary root bridge for VLANs 10, 20, 30, 40, and 50. Interfaces FastEthernet0/23 and FastEthernet0/24 were also grouped into Port-channel1 using LACP.

```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#spanning-tree mode rapid-pvst
SW1(config)#spanning-tree vlan 10,20,30,40,50 root primary
SW1(config)#
SW1(config)#interface range FastEthernet0/23 - 24
SW1(config-if-range)#description ETHERCHANNEL_TO_SW2
SW1(config-if-range)#switchport mode trunk
SW1(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50
SW1(config-if-range)#channel-group 1 mode active
SW1(config-if-range)#no shutdown
SW1(config-if-range)#exit
SW1(config)#
SW1(config)#interface Port-channel1
SW1(config-if)#description ETHERCHANNEL_TO_SW2
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
```

Figure 4.7.1 - Rapid STP and EtherChannel configuration on SW1

Figure 4.7.1 shows SW1 configured as the primary root bridge and the EtherChannel link toward SW2. Port-channel1 is configured as a trunk and carries VLANs 10, 20, 30, 40, and 50.

```
SW1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     00E0.A398.D254
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     00E0.A398.D254
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Po1              Desg FWD 12        128.27   Shr

SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

Port        Vlans allowed and active in management domain
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,40,50
Fa0/1       10,20,30,40,50
Fa0/2       10,20,30,40,50

SW1#
```

Figure 4.7.2 - Verification of Rapid STP and EtherChannel on SW1

Figure 4.7.2 confirms that SW1 is the root bridge for the VLAN being verified. The show etherchannel summary command also indicates that logical channel Po1 has formed correctly, with FastEthernet0/23 and FastEthernet0/24 shown as active EtherChannel member ports. The output demonstrates Rapid STP operation and confirms the presence of logical channel Po1. Port-channel1 is active, and the physical links are included in the EtherChannel through LACP. Trunk verification also shows the VLANs permitted on the links between the switches.

A complementary configuration was implemented on SW2. It was configured as the secondary root bridge for the VLANs in use and configured with the same EtherChannel link toward SW1.

```
SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#
SW2(config)#spanning-tree mode rapid-pvst
SW2(config)#spanning-tree vlan 10,20,30,40,50 root secondary
SW2(config)#
SW2(config)#interface range FastEthernet0/23 - 24
SW2(config-if-range)#description ETHERCHANNEL_TO_SW1
SW2(config-if-range)#switchport mode trunk
SW2(config-if-range)#switchport trunk allowed vlan 10,20,30,40,50
SW2(config-if-range)#channel-group 1 mode active
SW2(config-if-range)#no shutdown
SW2(config-if-range)#exit
SW2(config)#
SW2(config)#interface Port-channel1
SW2(config-if)#description ETHERCHANNEL_TO_SW1
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk allowed vlan 10,20,30,40,50
SW2(config-if)#no shutdown
SW2(config-if)#exit
SW2(config)#
SW2(config)#end
SW2#write memory
```

Figure 4.7.3 - Rapid STP and EtherChannel configuration on SW2

Figure 4.7.3 shows SW2 configured as the secondary root bridge and interfaces FastEthernet0/23 and FastEthernet0/24 assigned to Port-channel1.

With this configuration, the connection between SW1 and SW2 becomes more resilient and efficient. EtherChannel provides a redundant logical link between the central switches, while Rapid STP prevents loops when multiple Layer 2 connections exist.

4.8 NETWORK SERVICE CONFIGURATION

After basic connectivity, VLANs, routing, and redundancy mechanisms were configured, the network services required to test the infrastructure were set up. The project uses Web, DNS, FTP, and E-mail services hosted in the Data Center, together with a separate Web server in the DMZ.

The DNS server was configured to resolve the domain names used in the project. This allows end stations to access services by name rather than only by IP address.

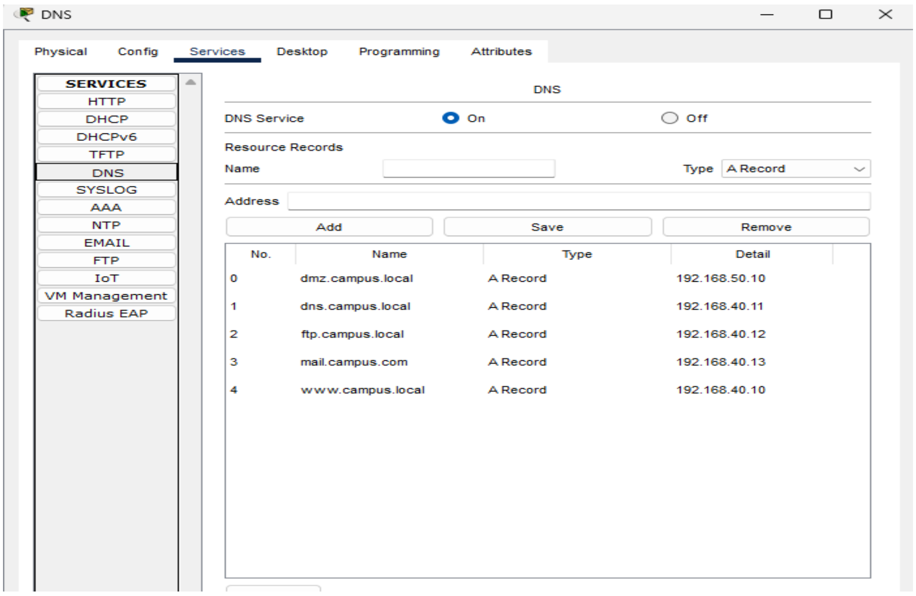

Figure 4.8.1 - DNS service configuration

The internal Web server was configured in the Data Center and is used to test HTTP access from permitted VLANs. The configuration consisted of enabling the HTTP service on the internal Web server, which is accessible through the domain www.campus.local.

An FTP server was configured to test file transfers. It is located in the Data Center, and access to it is controlled through the security policies.

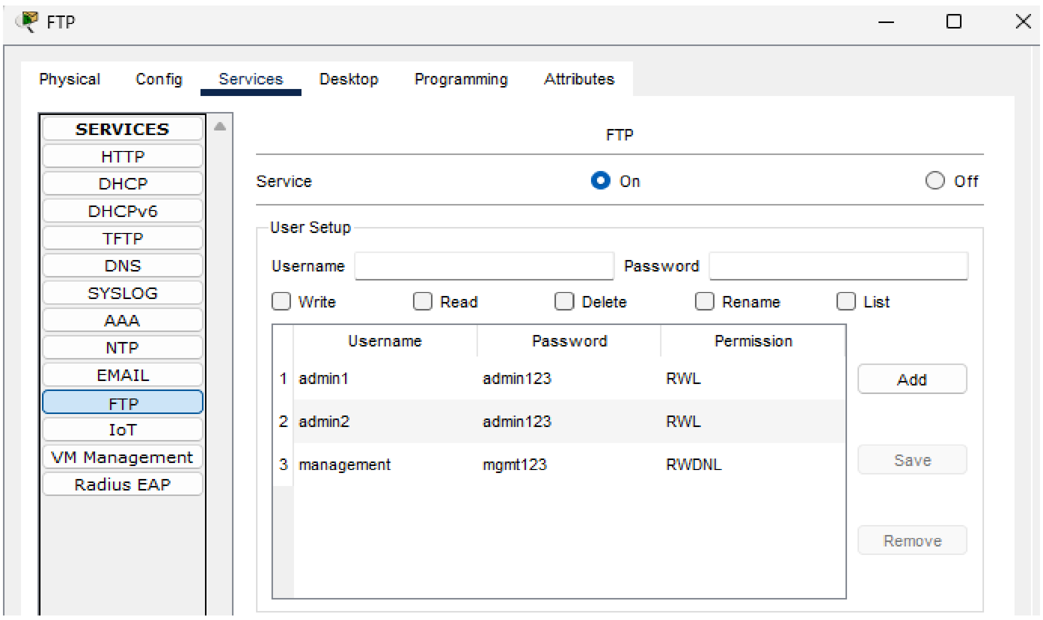

Figure 4.8.2 - FTP service configuration and user accounts used to test server access

The E-mail service was configured on the dedicated server in the Data Center. The domain campus.com was used for this service.

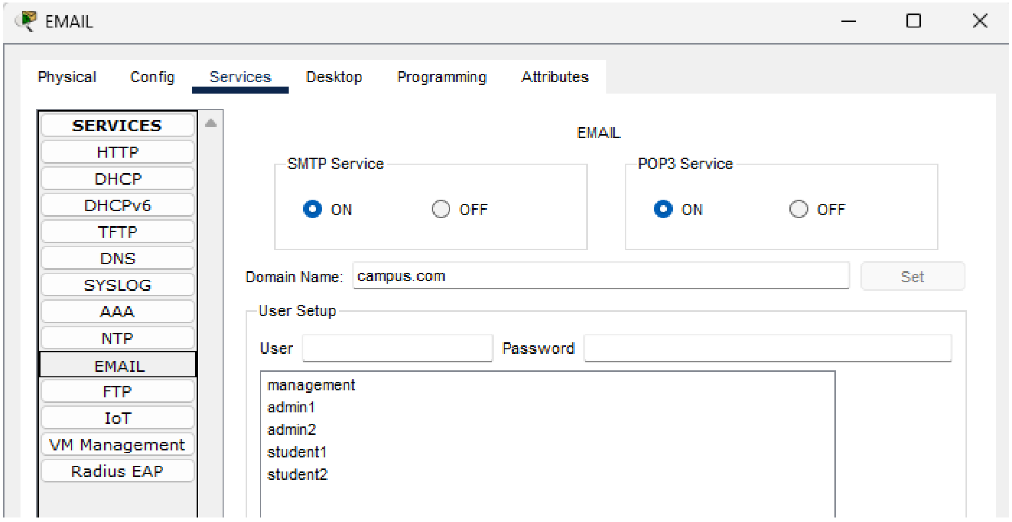

Figure 4.8.3 - E-mail service configuration, showing the e-mail domain and the accounts used for communication between users

Separate from the internal servers, the WEB-DMZ server was configured in the DMZ by enabling its HTTP service. It represents a Web service exposed in a controlled manner and separated from the Data Center, and it can be accessed through the domain dmz.campus.local.

4.9 ACL POLICY CONFIGURATION

Access control lists were configured to control traffic between VLANs. The ACLs were applied to the VLAN interfaces of the Layer 3 switches according to the area from which the traffic originates.

The first ACL policy was configured for the Students VLAN. This area can access only the required services, such as DNS, the internal Web server, and E-mail. Access to the management area, FTP server, and DMZ is restricted.

```
SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#no ip access-list extended VLAN30_IN
SW1(config)#
SW1(config)#ip access-list extended VLAN30_IN
SW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
SW1(config-ext-nacl)#permit udp 192.168.30.0 0.0.0.255 host 192.168.40.11 eq 53
SW1(config-ext-nacl)#permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.11 eq 53
SW1(config-ext-nacl)#permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.10 eq 80
SW1(config-ext-nacl)#permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.10 eq 443
SW1(config-ext-nacl)#permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.13 eq 25
SW1(config-ext-nacl)#permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.13 eq 110
SW1(config-ext-nacl)#deny tcp 192.168.30.0 0.0.0.255 host 192.168.40.12 eq 21
SW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
SW1(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
SW1(config-ext-nacl)#permit ip any any
SW1(config-ext-nacl)#exit
SW1(config)#
SW1(config)#interface Vlan30
SW1(config-if)#ip access-group VLAN30_IN in
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
```

Figure 4.9.1 - ACL configuration for the Students VLAN

A more permissive policy was configured for the Admin VLAN because this area has an administrative role. Users in this VLAN can access the internal servers and the Web service in the DMZ, while uncontrolled access to the DMZ remains restricted.

```
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#no ip access-list extended VLAN20_IN
SW1(config)#
SW1(config)#ip access-list extended VLAN20_IN
SW1(config-ext-nacl)#permit ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
SW1(config-ext-nacl)#permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq 80
SW1(config-ext-nacl)#permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq 443
SW1(config-ext-nacl)#deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
SW1(config-ext-nacl)#permit ip any any
SW1(config-ext-nacl)#exit
SW1(config)#
SW1(config)#interface Vlan20
SW1(config-if)#ip access-group VLAN20_IN in
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
SW1#
```

Figure 4.9.2 - ACL configuration for the Admin VLAN

An isolation policy was configured for the DMZ to separate it from the internal network. The WEB-DMZ server is kept separate from the internal servers, and traffic initiated from this area toward the internal VLANs is restricted.

```
SW1#enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#
SW1(config)#no ip access-list extended DMZ_IN
SW1(config)#
SW1(config)#ip access-list extended DMZ_IN
SW1(config-ext-nacl)#permit tcp 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255 established
SW1(config-ext-nacl)#permit udp 192.168.50.0 0.0.0.255 host 192.168.40.11 eq 53
SW1(config-ext-nacl)#permit tcp 192.168.50.0 0.0.0.255 host 192.168.40.11 eq 53
SW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
SW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
SW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
SW1(config-ext-nacl)#deny ip 192.168.50.0 0.0.0.255 192.168.40.0 0.0.0.255
SW1(config-ext-nacl)#permit ip any any
SW1(config-ext-nacl)#exit
SW1(config)#
SW1(config)#interface Vlan50
SW1(config-if)#ip access-group DMZ_IN in
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#write memory
Building configuration...
[OK]
SW1#
```

Figure 4.9.3 - ACL configuration for the DMZ

With these policies configured, traffic between the network's functional areas is controlled according to the role of each VLAN. The infrastructure therefore permits access to required services while restricting unauthorized communication toward sensitive areas.

4.10 FINAL IMPLEMENTED TOPOLOGY

After configuring the VLANs, inter-VLAN routing, redundancy mechanisms, network services, and ACL policies, the final topology represents a fully functional Data Center-Campus infrastructure. Figure 4.10.1 shows the final network structure, divided into distinct functional areas and organized according to the role of each segment.

The topology highlights the Routing Layer, Core / Distribution, Campus Distribution, Campus Access, Data Center, and DMZ areas. It also identifies the VLANs used in the project: VLAN 10 for Management, VLAN 20 for Admin, VLAN 30 for Students, VLAN 40 for Internal Servers, and VLAN 50 for the DMZ. This overview makes it possible to see how the network areas are interconnected and logically separated through VLANs.

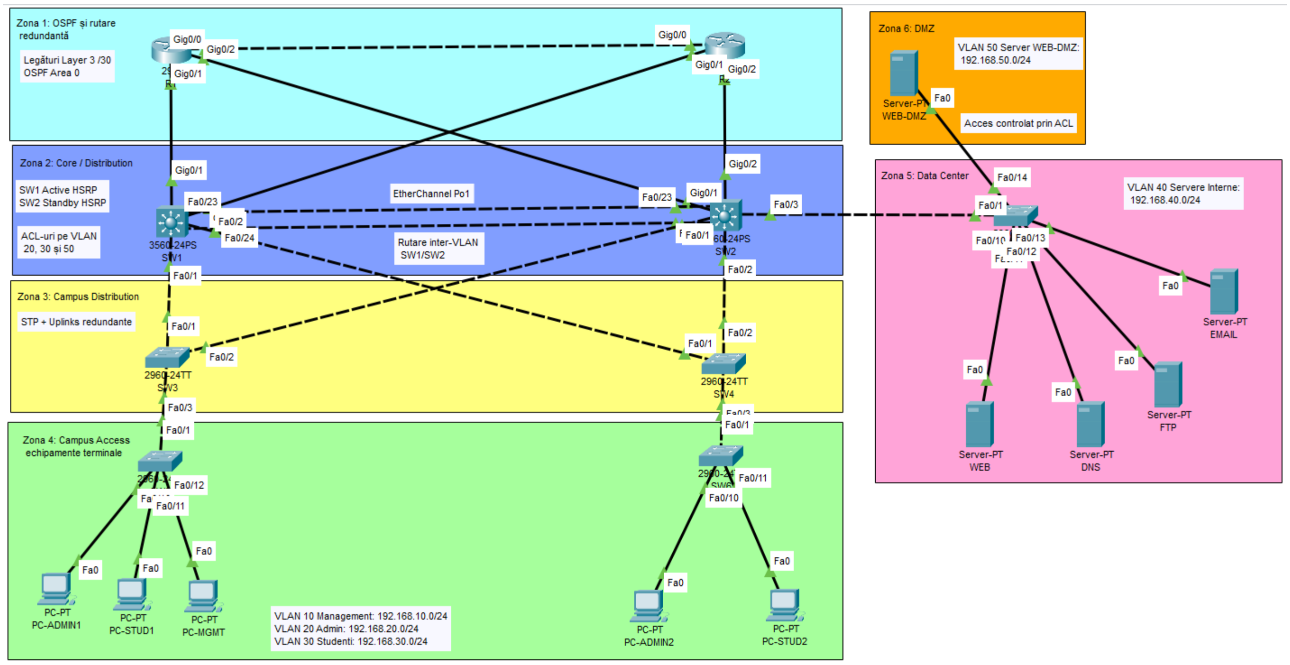

Figure 4.10.1 - Final implemented topology highlighting the functional areas and VLANs used

5. NETWORK TESTING AND VALIDATION

5.1 BASIC CONNECTIVITY TESTING

After the main configurations were completed, the infrastructure's basic connectivity was tested. The purpose of these tests was to verify communication between the Layer 3 devices, VLAN gateways, end stations, and internal servers.

The first test was performed from router R1 toward the other Layer 3 devices in the routing and Core / Distribution areas. Connections to R2, SW1, and SW2 were tested.

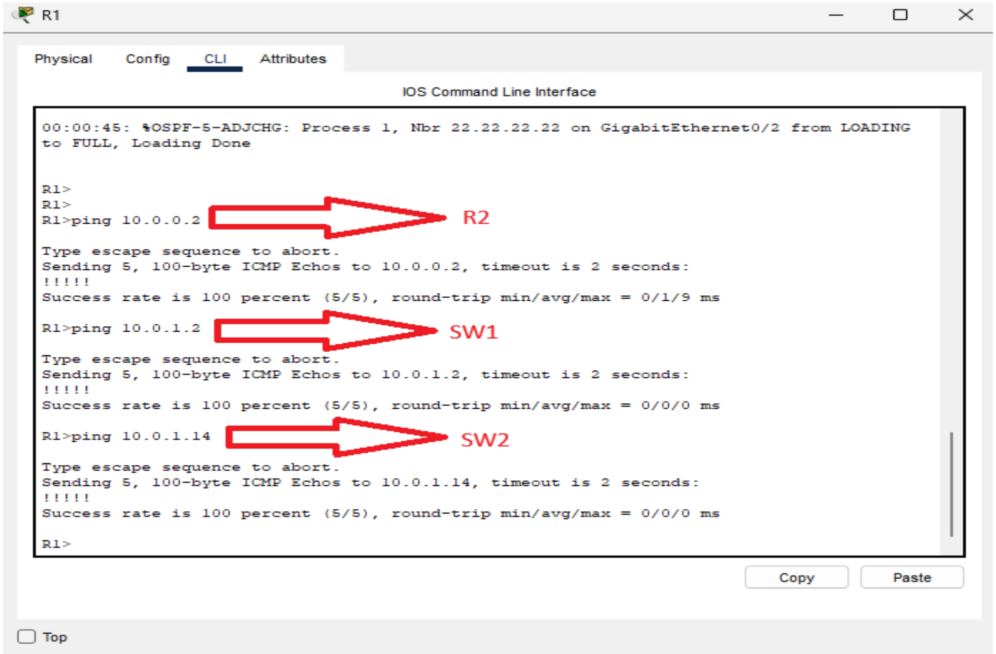

Figure 5.1.1 - Layer 3 connectivity testing from router R1

The next test verified access to the Admin VLAN gateway. The station sent ICMP packets to address 192.168.20.1, which represents the virtual gateway configured for VLAN 20.

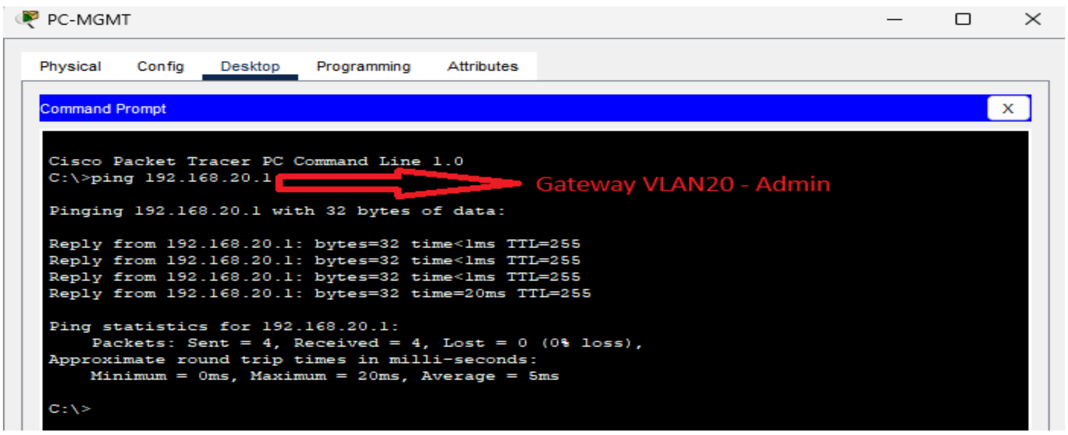

Figure 5.1.2 - Connectivity test to the VLAN 20 gateway

Figure 5.1.2 shows that the Admin VLAN gateway responds correctly to the ping test. This result confirms that the HSRP virtual address is active and can be used as the gateway for that VLAN.

Connectivity from the station in the Students VLAN to its own gateway, 192.168.30.1, was also tested.

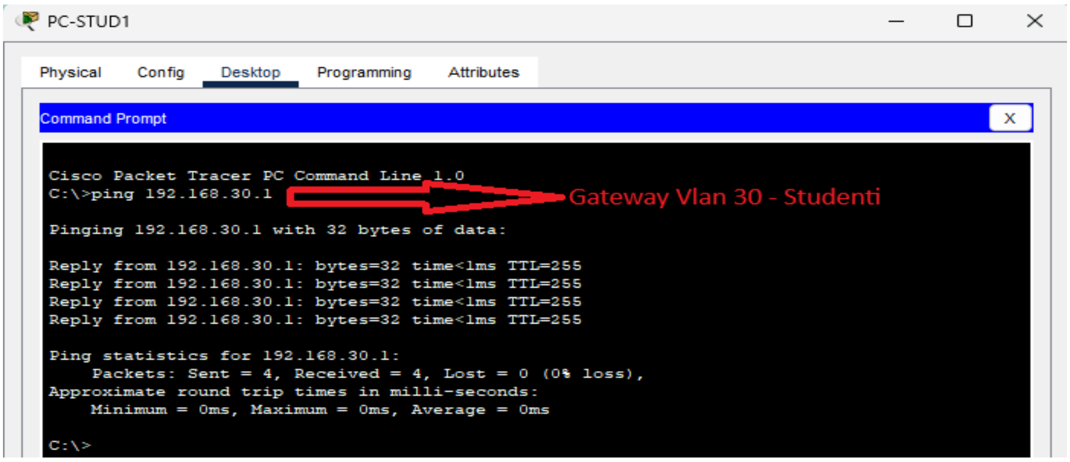

Figure 5.1.3 - Connectivity test to the VLAN 30 gateway

To verify communication between the VLANs and the Data Center area, access from PC-ADMIN1 to the internal Web server was tested. Communication with a station in the Students VLAN was also tested.

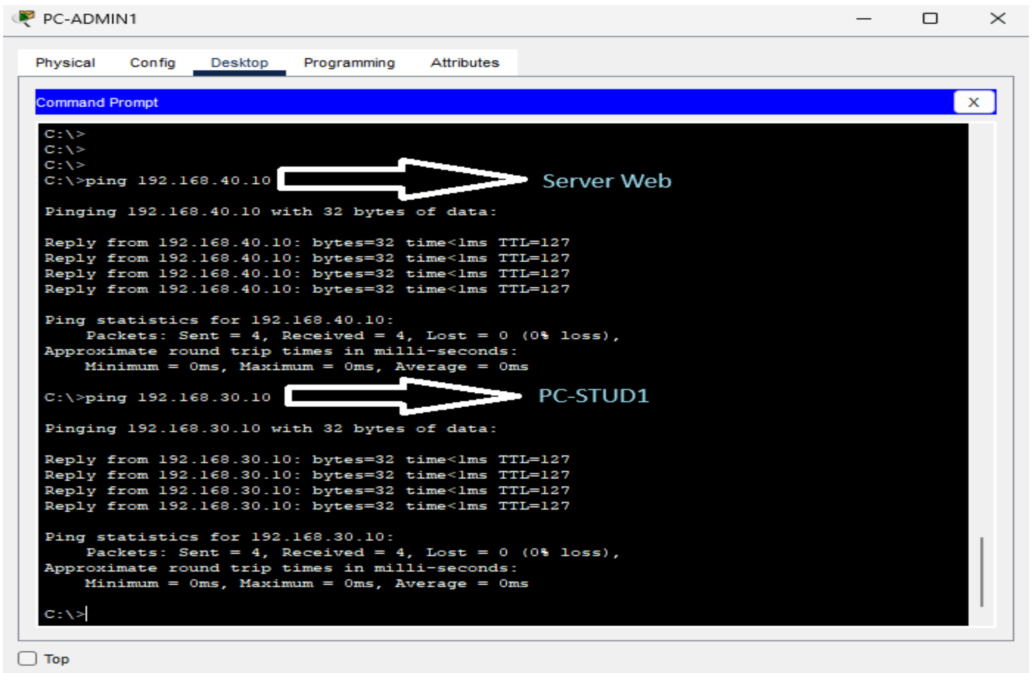

Figure 5.1.4 - Connectivity testing between the Admin VLAN, Data Center, and Students VLAN

Figure 5.1.4 shows that PC-ADMIN1 can communicate both with the internal Web server in the Data Center and with a station in the Students VLAN. This confirms the operation of inter-VLAN routing and connectivity between the network's main areas.

The results show that the infrastructure has functional basic connectivity. The Layer 3 devices communicate with one another, the VLAN gateways respond correctly, and traffic between the Campus and Data Center areas can be forwarded according to the implemented configuration.

5.2 NETWORK SERVICE TESTING

The first services tested were DNS and the Web service hosted in the Data Center, with the Web service used to verify HTTP access. DNS resolves the domain names configured in the project, including www.campus.local, ftp.campus.local, dmz.campus.local, and mail.campus.com.

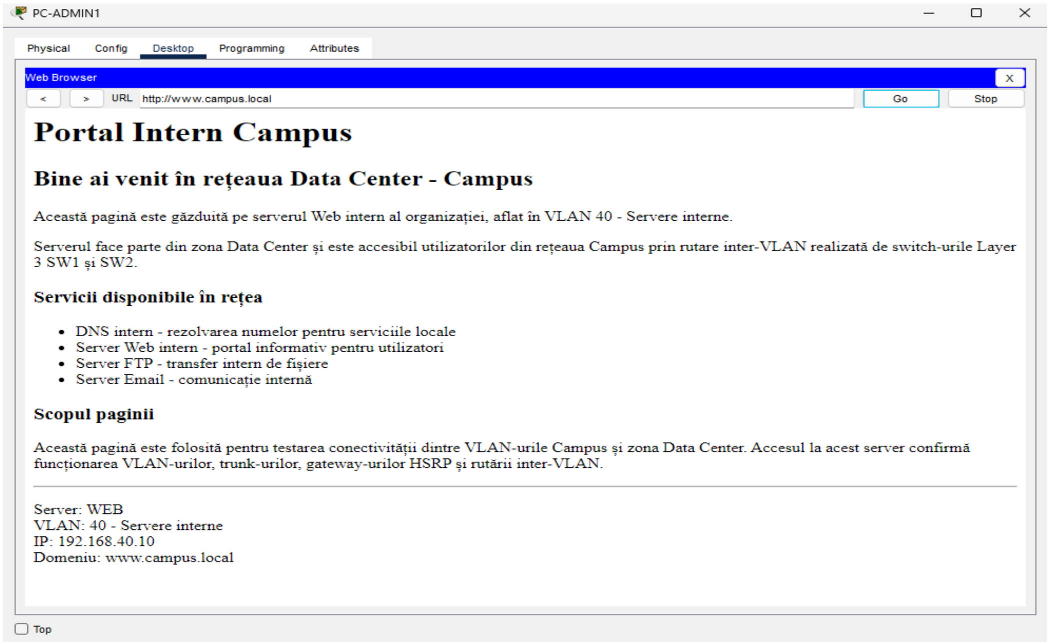

Figure 5.2.1 - Testing DNS resolution for network services and access to the internal Web server

The FTP service was also tested by connecting to the server ftp.campus.local. This test verifies file-transfer functionality and access to the FTP server in the Data Center.

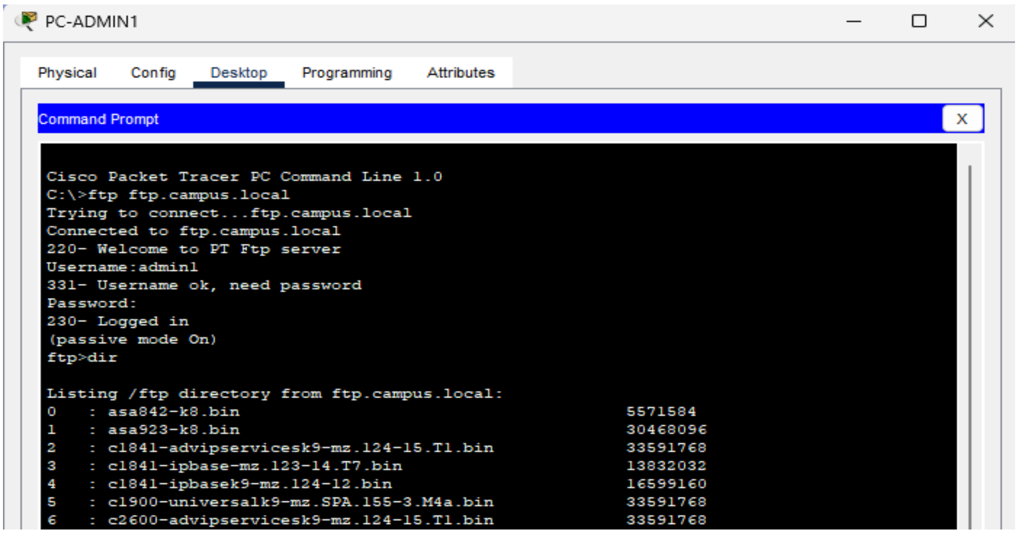

Figure 5.2.2 - FTP service access test

For the E-mail service, accounts configured under the campus.com domain were used. The test consisted of sending and receiving a message between two end stations.

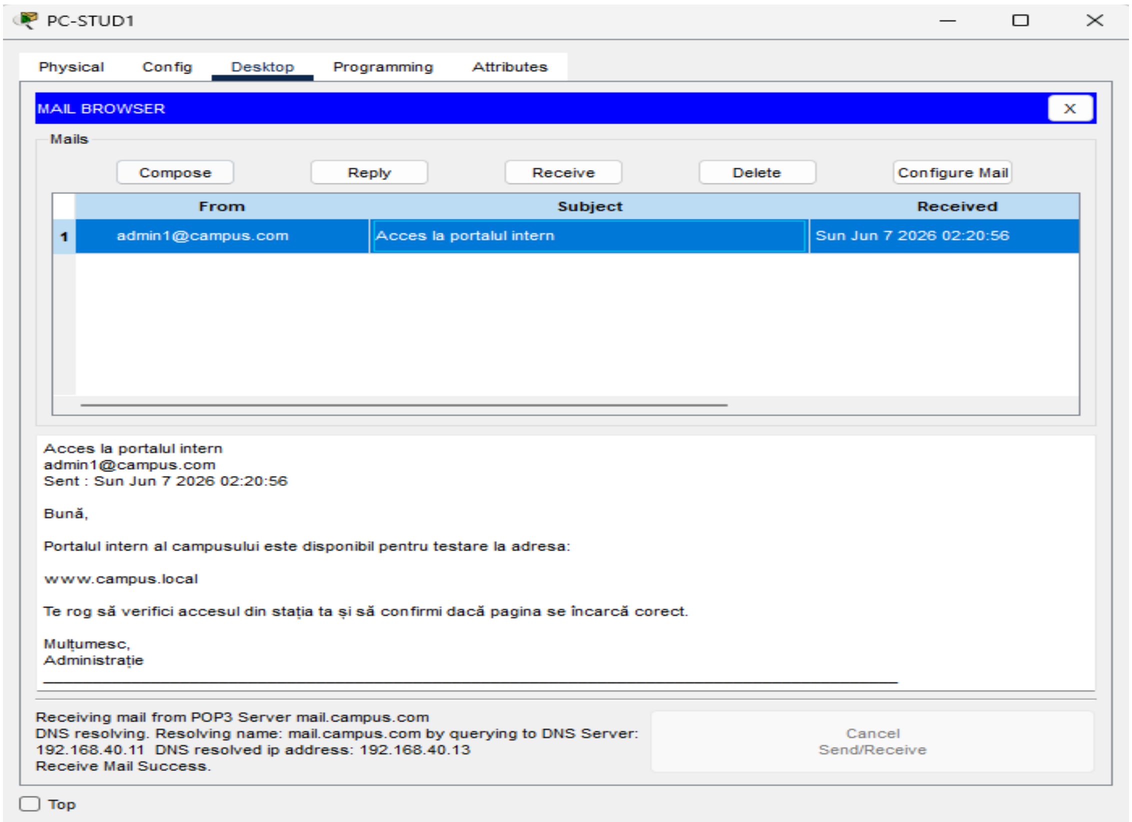

Figure 5.2.3 - E-mail service test

Figure 5.2.3 confirms the operation of the e-mail service by showing a message transmitted between users configured in the network.

Separate from the internal services, the Web server in the DMZ was also tested. It was accessed through the domain dmz.campus.local according to the access policies defined in the project.

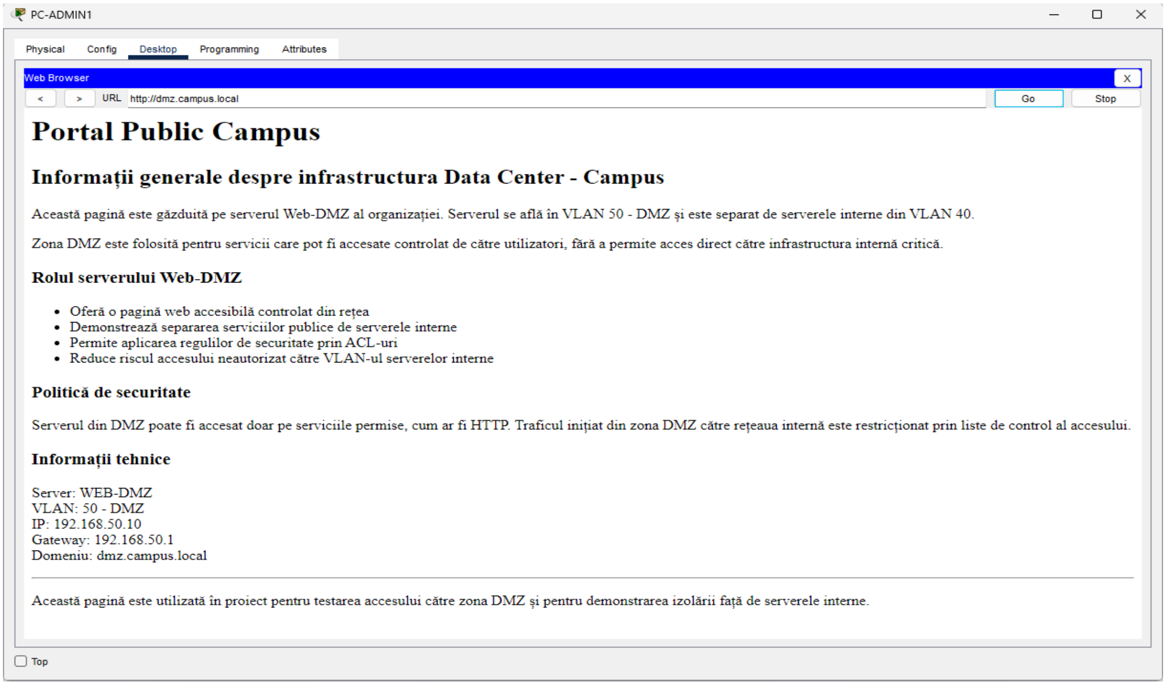

Figure 5.2.4 - Access test for the Web server in the DMZ

The test results show that the configured network services operate correctly. End stations can access the required resources, and the servers in the Data Center and DMZ respond according to the implemented configuration.

5.3 SECURITY POLICY TESTING

After the network services were tested, the security policies applied through ACLs were verified. The purpose of these tests was to confirm that access between VLANs is permitted or restricted according to the rules defined in the project.

For the Students VLAN, both permitted and restricted communications were tested. Station PC-STUD1 can access required services such as the internal Web server, DNS, and E-mail, but it should not have access to the FTP server, the DMZ, or the Management VLAN.

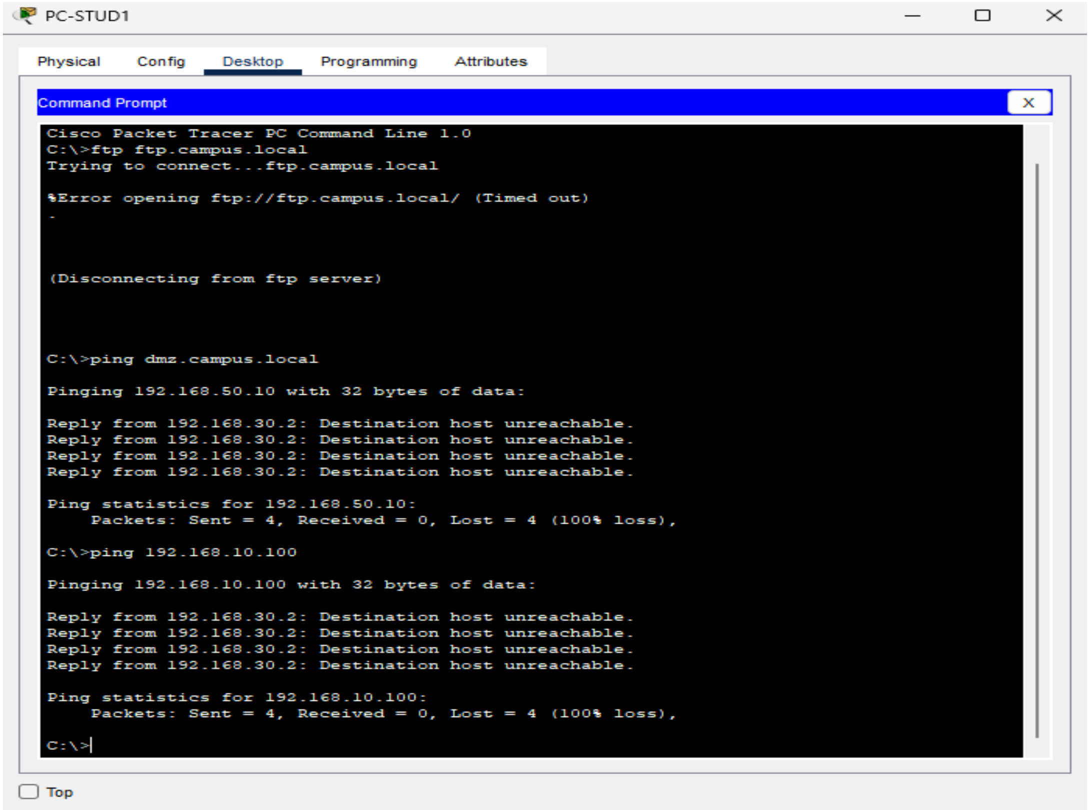

Figure 5.3.1 - Restricted-access testing from the Students VLAN

Figure 5.3.1 shows that access to restricted resources is blocked. This result confirms that the ACL policy for the Students VLAN is applied correctly.

For the Admin VLAN, the tests verified access to the internal servers and to the Web service in the DMZ. Administrative users have a broader level of access, but traffic toward the DMZ remains controlled.

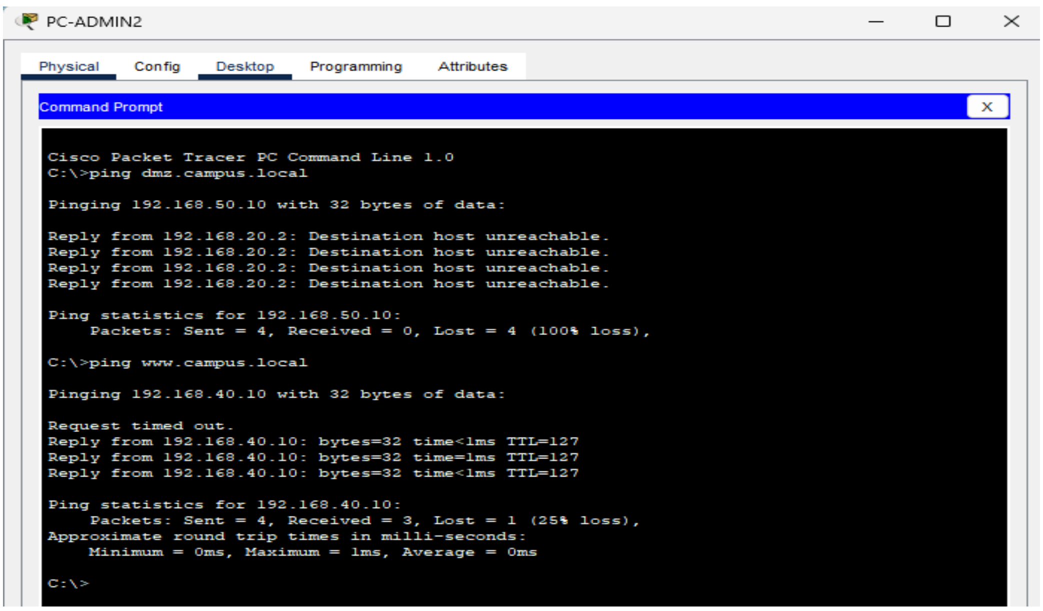

Figure 5.3.2 - Controlled-access testing from the Admin VLAN

Figure 5.3.2 confirms that users in the Admin VLAN can access only the Web service published in the DMZ and do not have direct access to other resources or segments within the DMZ. Administrative users have a broader level of access, but traffic toward the DMZ remains controlled.

The isolation of the DMZ from the internal network was also verified. The WEB-DMZ server is separated from the internal servers and must not provide direct access to sensitive areas of the infrastructure.

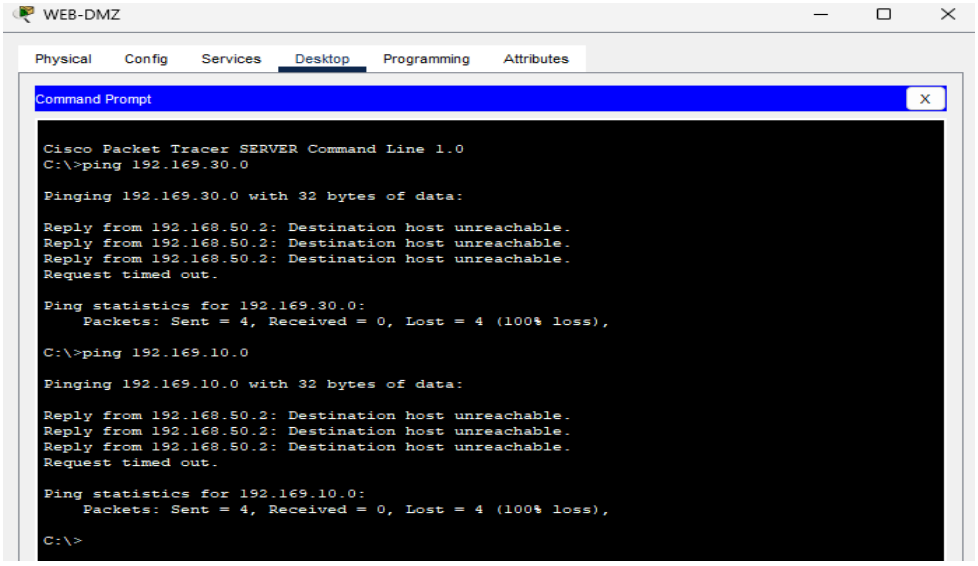

Figure 5.3.3 - DMZ isolation test

Figure 5.3.3 shows that unauthorized traffic from the DMZ toward internal networks is restricted. The DMZ therefore remains separated from the internal network infrastructure.

The test results confirm that the configured security policies operate correctly. The ACLs permit access to the required services and block unauthorized traffic toward protected areas, helping isolate important infrastructure segments.

5.4 REDUNDANCY MECHANISM TESTING

After connectivity and security policies were verified, the redundancy mechanisms configured in the topology were tested. The purpose of these tests was to confirm that the network can maintain communication if a link fails or an important device becomes unavailable.

5.4.1 HSRP

The first mechanism tested was HSRP, which provides redundant gateways for the VLANs. The test verified the transfer of the active role from SW1 to SW2 if the primary device becomes unavailable.

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface vlan 20
SW1(config-if)#shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to down

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#
```

Figure 5.4.1 - Simulating unavailability of the active gateway on SW1

```
SW2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100 P Standby  192.168.10.2    local           192.168.10.1
Vl20        20   100 P Active   local           unknown         192.168.20.1
Vl30        30   100 P Standby  192.168.30.2    local           192.168.30.1
Vl40        40   100 P Standby  192.168.40.2    local           192.168.40.1
Vl50        50   100 P Standby  192.168.50.2    local           192.168.50.1
SW2#
```

Figure 5.4.2 - SW2 assuming the Active role for VLAN 20

```
PC-ADMIN1
Physical   Config   Desktop   Programming   Attributes
Command Prompt

C:\>
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
Reply from 192.168.40.10: bytes=32 time=1ms TTL=125
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=7ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

Figure 5.4.3 - Connectivity verification after HSRP failover.

Figure 5.4.3 confirms that connectivity is maintained after the HSRP role change. Although one packet is temporarily lost during the transition, communication resumes and the VLAN 20 gateway responds correctly.

5.4.2 OSPF

To test redundancy through OSPF, a Layer 3 link failure between router R1 and switch SW1 was simulated. The purpose of the test was to verify OSPF's ability to recalculate the path and maintain communication through an alternative route.

```
R1>ping 192.168.40.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface GigabitEthernet0/1
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

00:02:26: %OSPF-5-ADJCHG: Process 1, Nbr 11.11.11.11 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Interface down or detached

R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    10.0.0.2        GigabitEthernet0/0
22.22.22.22       1   FULL/DR         00:00:35    10.0.1.6        GigabitEthernet0/2
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.0.1.8 [110/2] via 10.0.0.2, 00:00:32, GigabitEthernet0/0
O       10.0.1.12 [110/2] via 10.0.0.2, 00:02:03, GigabitEthernet0/0
                  [110/2] via 10.0.1.6, 00:02:03, GigabitEthernet0/2
O    192.168.10.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.20.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.30.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.40.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.50.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2

R1#
```

Figure 5.4.2.1 - Simulating a Layer 3 link failure between R1 and SW1 and verifying OSPF routes after the link failure

Figure 5.4.2.1 shows the shutdown of interface GigabitEthernet0/1 on R1, which is used for the connection to SW1, and the routing-information update after the link failure. The network continues to have routes to the internal segments through the remaining Layer 3 connections.

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:35    10.0.0.2        GigabitEthernet0/0
22.22.22.22       1   FULL/DR         00:00:35    10.0.1.6        GigabitEthernet0/2
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.0.1.8 [110/2] via 10.0.0.2, 00:00:32, GigabitEthernet0/0
O       10.0.1.12 [110/2] via 10.0.0.2, 00:02:03, GigabitEthernet0/0
                  [110/2] via 10.0.1.6, 00:02:03, GigabitEthernet0/2
O    192.168.10.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.20.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.30.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.40.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2
O    192.168.50.0 [110/2] via 10.0.1.6, 00:00:32, GigabitEthernet0/2

R1#ping 192.168.40.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

Figure 5.4.2.2 - Connectivity verification after OSPF route recalculation.

Figure 5.4.2.2 confirms that traffic to the internal server remains functional after the R1-SW1 link is interrupted. This result demonstrates that OSPF can use an alternative path within the infrastructure.

5.4.3 ETHERCHANNEL

To test EtherChannel redundancy, one of the two physical links between SW1 and SW2 was interrupted. The purpose of the test was to verify that logical link Port-channel1 remains operational through the remaining active physical connection.

```
SW1>show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
SW1>
```

Figure 5.4.3.1 - EtherChannel verification before a physical link is interrupted.

Figure 5.4.3.1 confirms that Port-channel1 is active and that interfaces FastEthernet0/23 and FastEthernet0/24 are included in the EtherChannel.

```
SW1>en
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/23
SW1(config-if)#shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/23(D) Fa0/24(P)
SW1#
```

Figure 5.4.3.2 - Simulating the failure of a physical EtherChannel member link

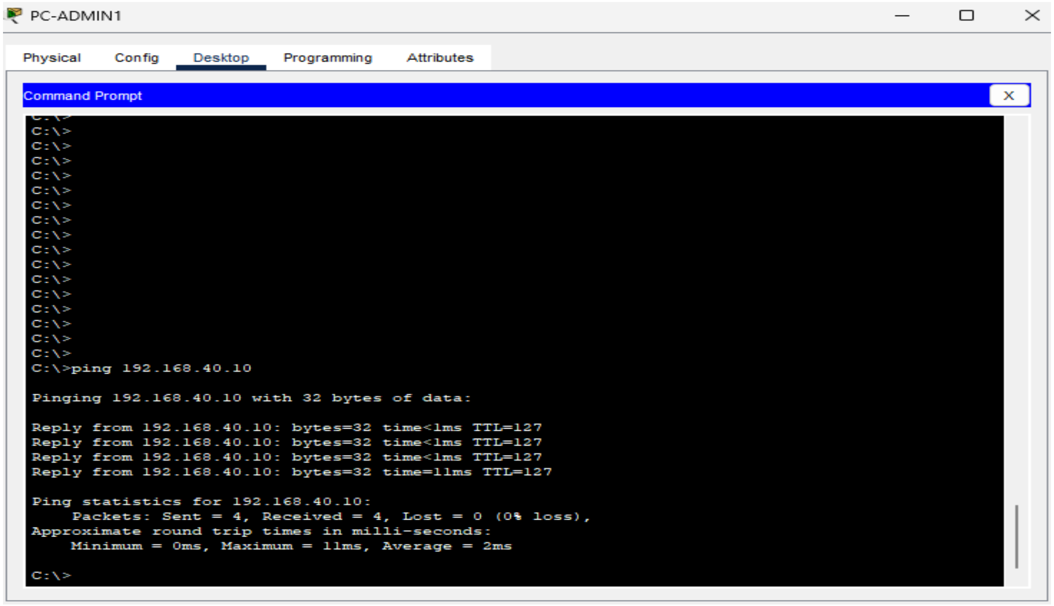

Figure 5.4.3.3 - Connectivity verification after EtherChannel failover

Figure 5.4.3.3 shows that station PC-ADMIN1 can continue to communicate with the internal Web server after one physical EtherChannel link is interrupted. This confirms that the logical link between SW1 and SW2 remains operational through the remaining active connection.

5.4.4 STP

To test Rapid STP, a Layer 2 link failure was simulated in the Campus Distribution area. The test was performed on switch SW4, which has redundant connections toward the Core / Distribution area.

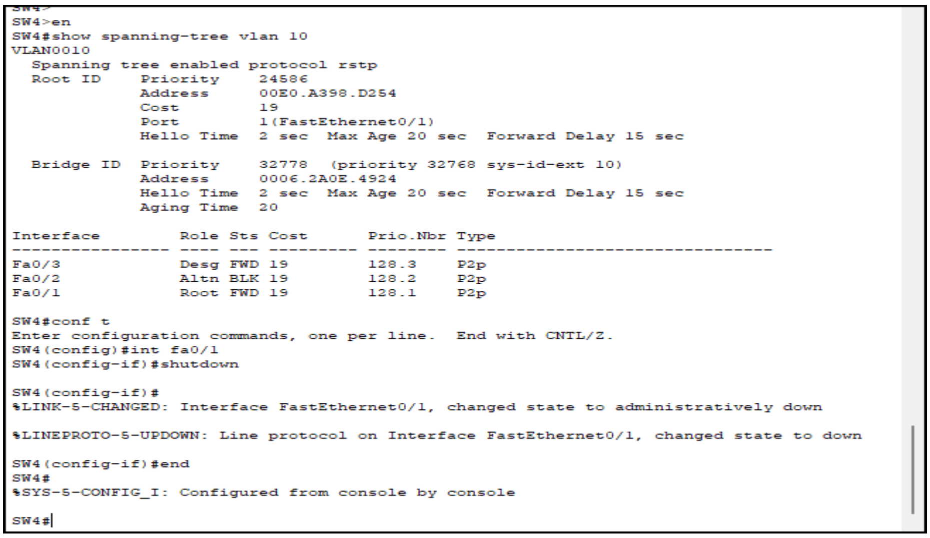

Figure 5.4.4.1 - Rapid STP verification before a link failure

Figure 5.4.4.1 shows the STP port states on SW4 before the topology change. Interface FastEthernet0/1 is the Root port in the Forwarding state, while interface FastEthernet0/2 is an alternate port in the Blocking state. The same capture simulates the link failure by shutting down interface FastEthernet0/1.

```
SW4(config-if)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console

SW4#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     00E0.A398.D254
             Cost        31
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0006.2A0E.4924
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/2            Root FWD 19        128.2    P2p

SW4#
```

Figure 5.4.4.2 - Rapid STP verification after the link failure

Figure 5.4.4.2 highlights the topology recalculation after interface FastEthernet0/1 is disabled. Interface FastEthernet0/2, which was previously an alternate port, becomes the Root port in the Forwarding state. This confirms that Rapid STP can activate an alternative path and maintain Layer 2 topology stability.

This test confirms that Rapid STP operates correctly when a redundant link changes state. The protocol prevents Layer 2 loops and enables an alternative link to be used when the primary link becomes unavailable.

6. CONCLUSIONS

The completed project demonstrates how a Data Center-Campus network infrastructure can be designed, configured, and tested in a simulation environment. The main objective of the thesis was to build an organized network capable of providing connectivity between users and services while also ensuring segmentation, availability, and security.

The topology was designed in Cisco Packet Tracer, which enabled the creation of a virtual infrastructure consisting of routers, switches, servers, and end stations. The environment made it possible to configure devices, test connectivity, and simulate normal-operation and redundancy scenarios without requiring physical equipment.

The network was divided into distinct functional areas, including the Routing Layer, Core / Distribution, Campus Distribution, Campus Access, Data Center, and DMZ. This organization clearly separated infrastructure roles and enabled more efficient traffic management. Through VLANs, users, servers, management devices, and the DMZ were logically separated according to the role of each area.

The implementation of inter-VLAN routing and OSPF provided communication between network segments and automatic route exchange between Layer 3 devices. At the same time, HSRP, EtherChannel, and Rapid STP increased network availability and helped maintain connectivity during simulated failures.

The Web, DNS, FTP, and E-mail services configured in the Data Center enabled testing of user access to internal resources. Separately, the Web server located in the DMZ demonstrated the role of an isolated area for services exposed in a controlled manner. Access to these services was managed through ACLs applied according to the role of each VLAN.

The tests confirmed the correct operation of the implemented configurations. Basic connectivity, service access, security policies, and redundancy mechanisms were validated. The results showed that the infrastructure can operate reliably and maintain communication when the state of links or devices changes.

In the future, the infrastructure could be extended by implementing a dedicated firewall between the main network areas to provide more advanced traffic and security-policy control. The project could also be expanded by adding NAT/PAT mechanisms, allowing external-network access and address management to be simulated in scenarios that more closely resemble real infrastructures.

Another possible development is the integration of centralized monitoring and management mechanisms such as SNMP and Syslog to track device status and identify network problems more quickly.

The project demonstrates the importance of a well-organized architecture in which segmentation, routing, redundancy, and security are considered together. The implemented topology shows that a Data Center-Campus network can be built modularly, provide controlled access to services, and remain operational in failover scenarios when the network technologies involved are configured correctly.

7. BIBLIOGRAPHY

[1] J. F. Kurose, K. W. Ross, Computer Networking: A Top-Down Approach, 8th ed. Pearson, 2021.

*[2] *** Cisco Systems, Campus LAN and Wireless LAN Solution Design Guide: https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html - accessed on 06.11.2025*

[3] *** Cisco Systems, “What is a data center?,” Cisco. : https://www.cisco.com/site/us/en/learn/topics/computing/what-is-a-data-center.html - accessed on 06.11.2025

[4] *** Cisco Systems, “Configure 802.1Q Trunking Between Catalyst Switches,” Cisco, Jun. 7, 2024. : https://www.cisco.com/c/en/us/support/docs/switches/catalyst-6000-series-switches/10599-88.html - accessed on 11.11.2025

[5] *** Cisco Press, “Inter-VLAN Routing using Layer 3 Switches,” Cisco Press, Jul. 29, 2020. : https://www.ciscopress.com/articles/article.asp?p=3089357&seqNum=6 - accessed on 09.03.2026

[6] *** Cisco Systems, “OSPF Configuration Guide - OSPF,” Cisco IOS XE 17, Cisco, Sep. 15, 2025. : https://www.cisco.com/c/en/us/td/docs/switches/lan/c9000/lyr3-fwd/ospf/ospf-configuration-guide/ospf.html. - accessed on 12.03.2026

[7] *** Cisco Systems, “Configuring Spanning Tree Protocol,” Layer 2 Configuration Guide, Cisco IOS XE 17.14.x, Cisco Catalyst 9200 Series Switches, Cisco, Apr. 5, 2024: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9200/software/release/17-14/configuration_guide/lyr2/b_1714_lyr2_9200_cg/configuring_spanning_tree_protocol.html - accessed on 25.03.2026

[8] *** Cisco Systems, “Configuring EtherChannels,” Layer 2 Configuration Guide, Cisco IOS XE 17.14.x, Cisco Catalyst 9200 Series Switches, Cisco, Apr. 5, 2024. : https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9200/software/release/17-14/configuration_guide/lyr2/b_1714_lyr2_9200_cg/configuring_etherchannels.html - accessed on 26.03.2026

[9] *** Cisco Systems, “FHRP Configuration Guide - HSRP,” Cisco IOS XE 17, Cisco, Sep. 15, 2025. : https://www.cisco.com/c/en/us/td/docs/switches/lan/c9000/lyr3-fwd/fhrp/fhrp-configuration-guide/hsrp.html. - accessed on 10.04.2026

[10] *** Cisco Systems, “ACLs Configuration Guide - Access Control Lists,” Cisco IOS XE 17, Cisco, Aug. 25, 2025. : https://www.cisco.com/c/en/us/td/docs/switches/lan/c9000/security/acls/acls-configuration-guide/access-control-lists.html - accessed on 10.04.2026

[11] *** Mozilla Developer Network, “Overview of HTTP,” MDN Web Docs. : https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview. - accessed on 13.04.2026

[12] *** Cisco Systems, “Understand DNS,” Cisco, Oct. 23, 2025. : https://www.cisco.com/c/en/us/support/docs/security/umbrella/225283-understand-dns.html - accessed on 15.04.2026

[13] *** GeeksForGeeks, “File Transfer Protocol (FTP) in Application Layer” 11 Jun, 2026 : https://www.geeksforgeeks.org/computer-networks/file-transfer-protocol-ftp-in-application-layer/ - accessed on 13.04.2026

[14] *** Cloudflare, “What is the Simple Mail Transfer Protocol (SMTP)?,”: https://www.cloudflare.com/learning/email-security/what-is-smtp/. - accessed on 29.04.2026

[15] *** GeeksForGeeks “What is a DMZ Network in CCNA?” 4 Oct, 2025 : https://www.geeksforgeeks.org/computer-networks/what-is-a-dmz-network-in-ccna/ - accessed on 10.05.2026

[16] *** Cisco Systems, “What is a Router?”, Cisco. : https://www.cisco.com/site/us/en/learn/topics/small-business/what-is-a-router.html - accessed on 11.05.2026

[17] *** Cisco Systems, “What Is Network Switching?”, Cisco. : https://www.cisco.com/site/us/en/learn/topics/networking/what-is-network-switching.html - accessed on 12.05.2026